

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-13 17:57:14

Data & configuration

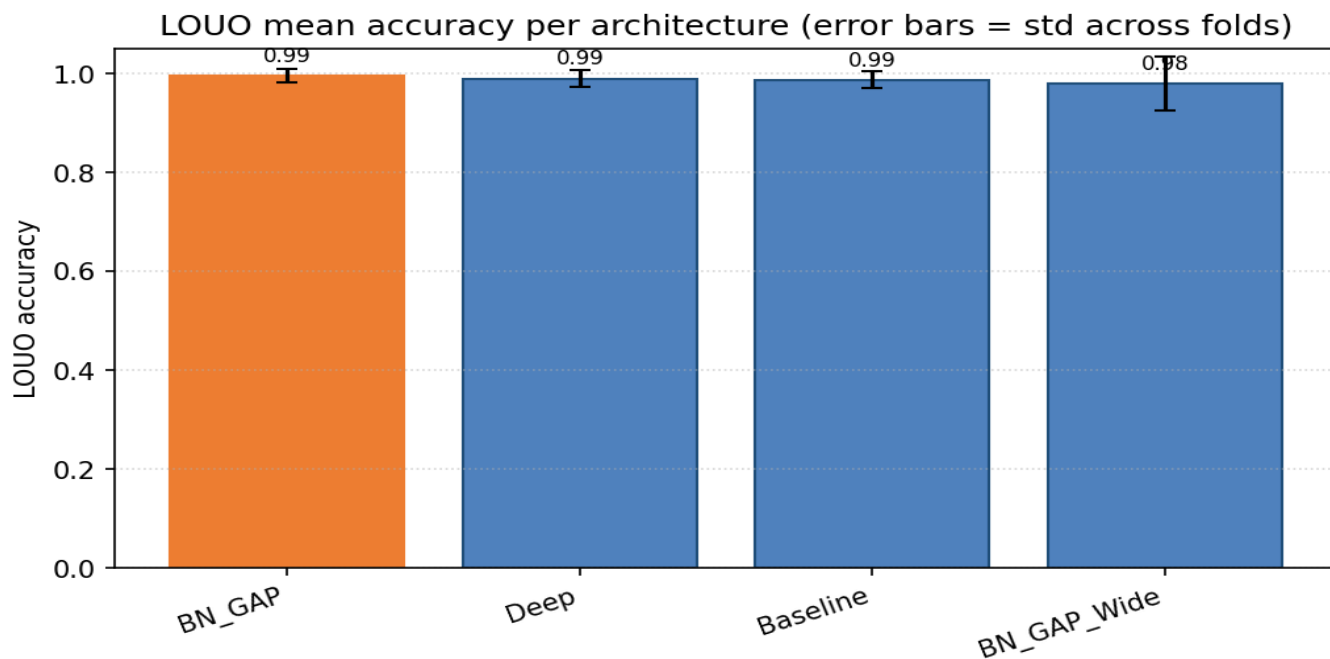
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	Double_Nothing
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	BN_GAP_Wide, Deep, Baseline, BN_GAP
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

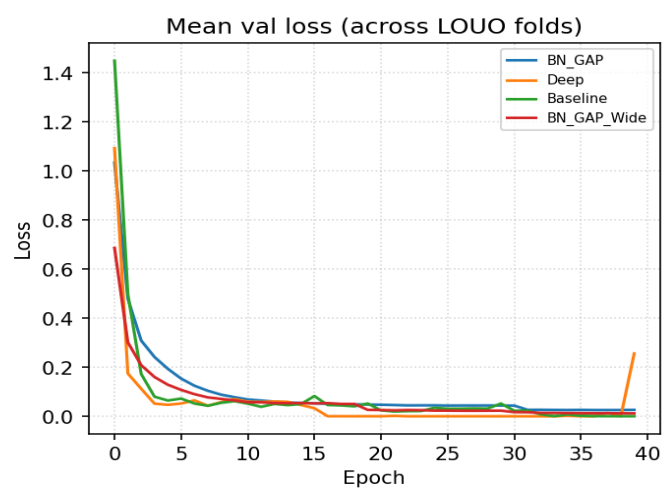
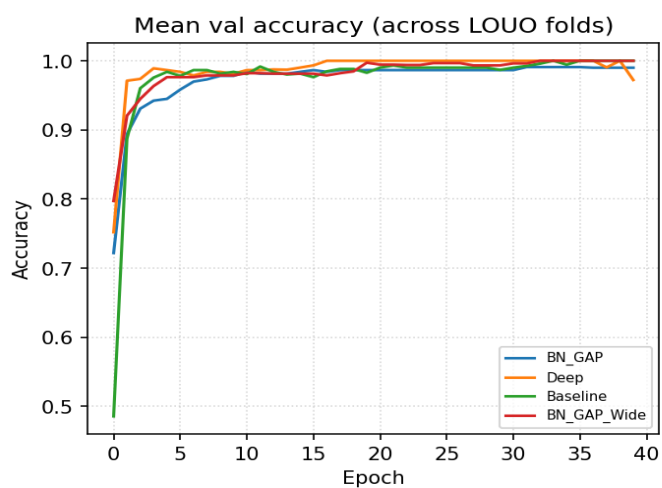
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP	57,190	0.9942	0.0136	13	—
Deep	203,878	0.9891	0.0165	13	—
Baseline	113,190	0.9865	0.0172	13	—
BN_GAP_Wide	104,294	0.9788	0.0553	13	—

Best LOUO mean: BN_GAP

LOUO accuracy per variant

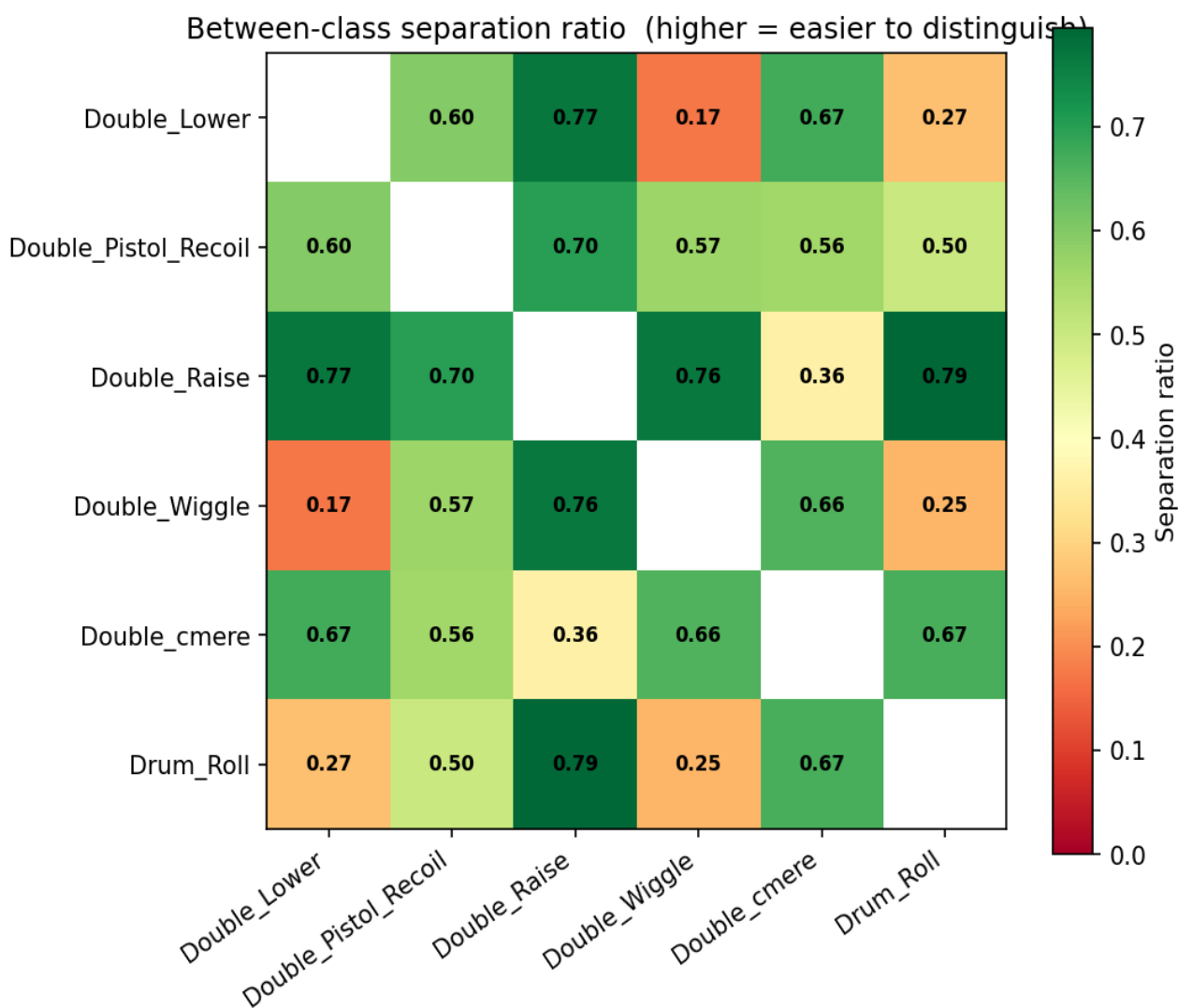
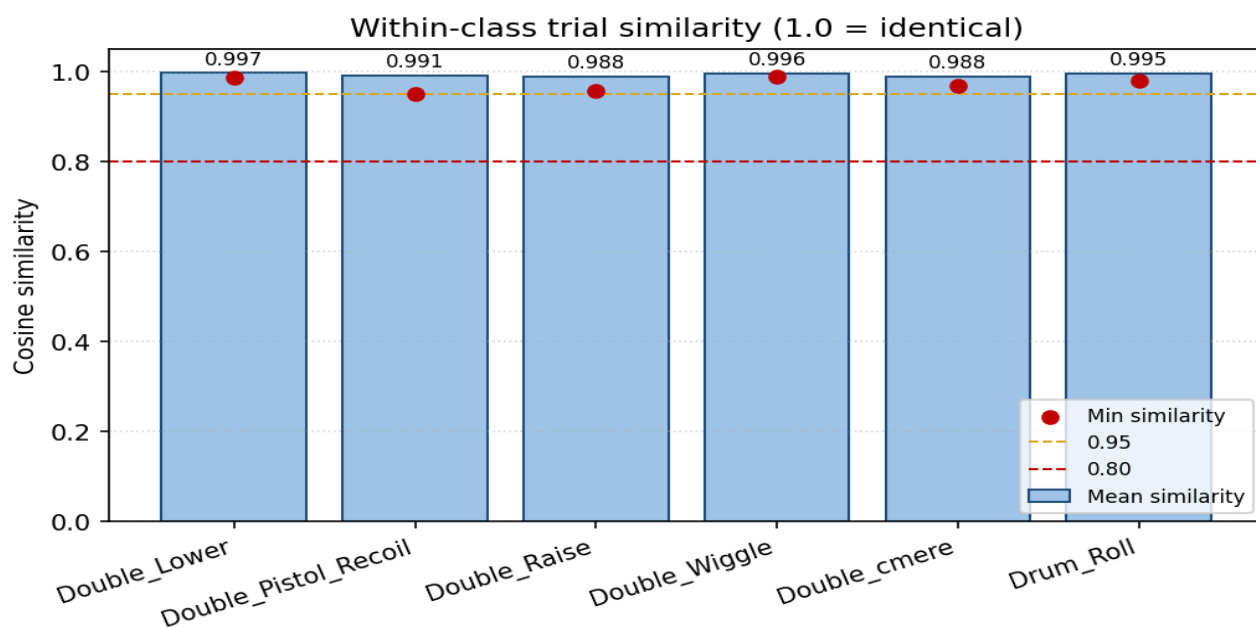


Mean training curves across LOUO folds

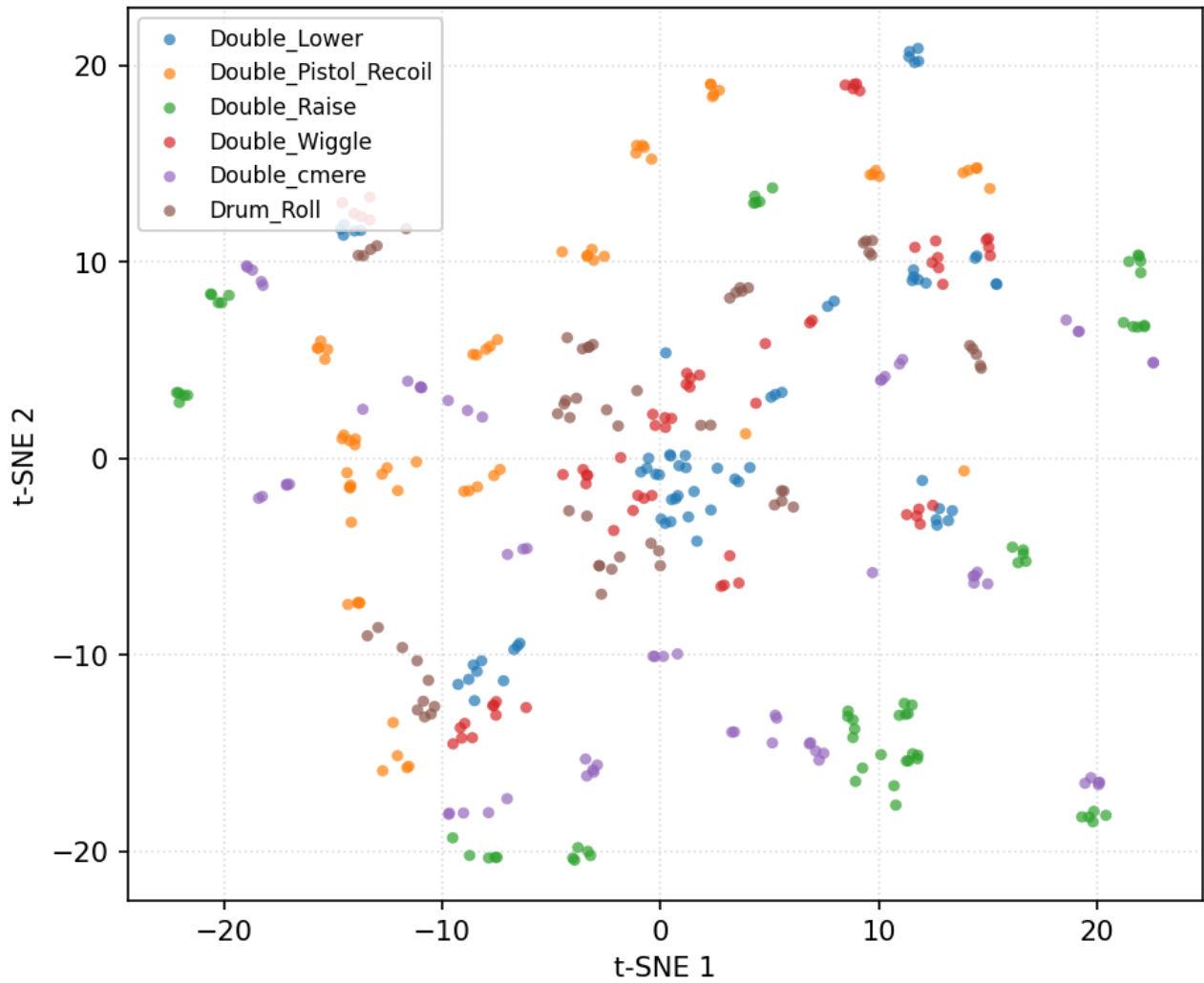


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



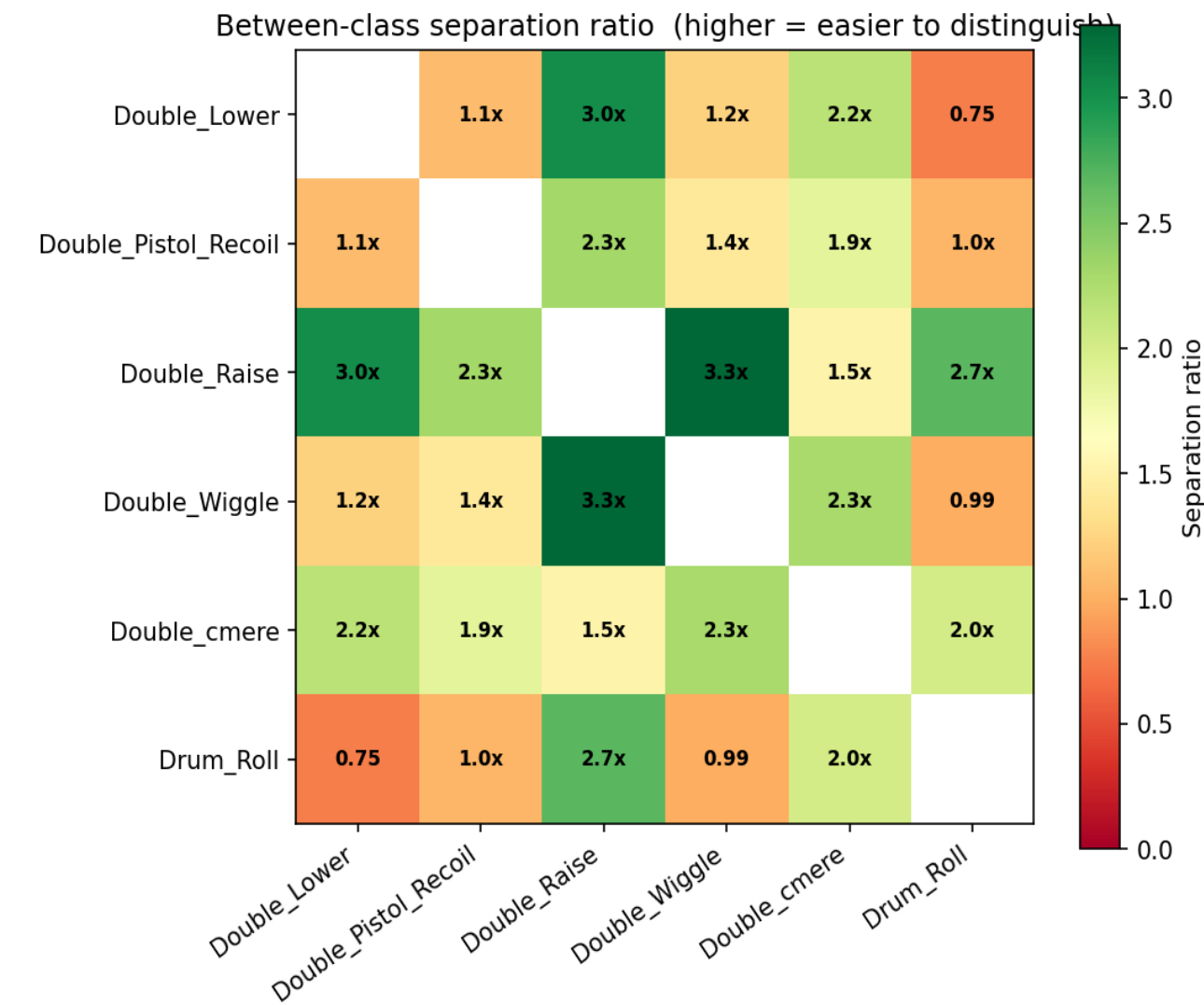
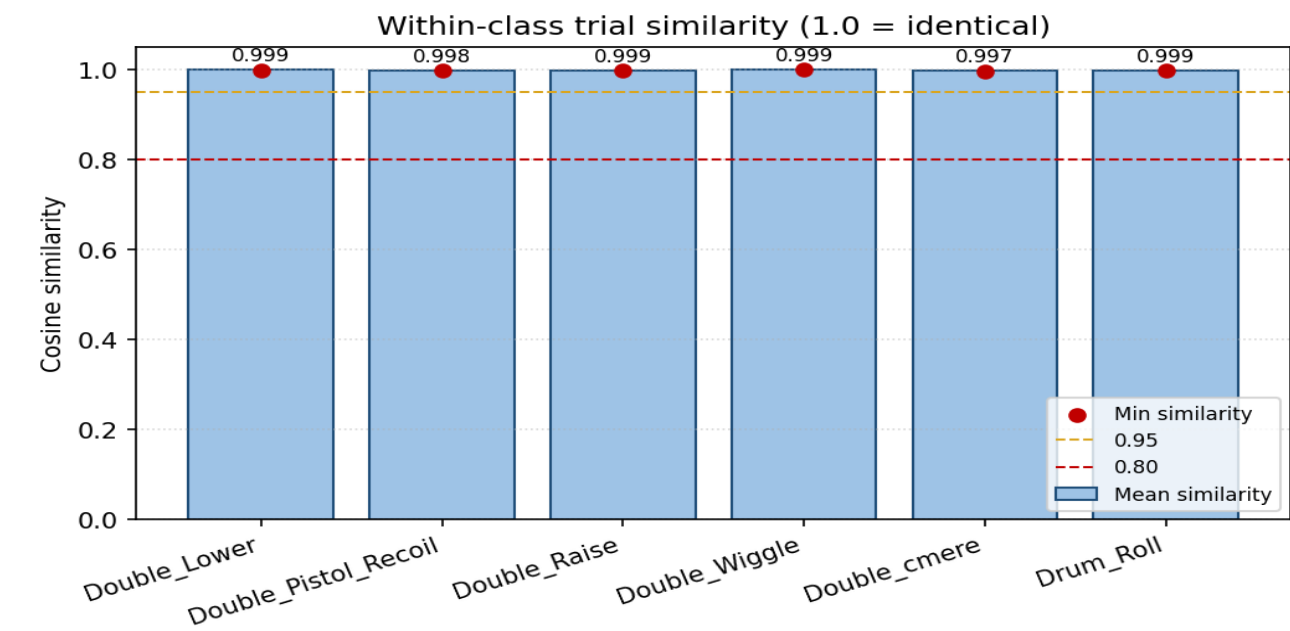
t-SNE feature space projection



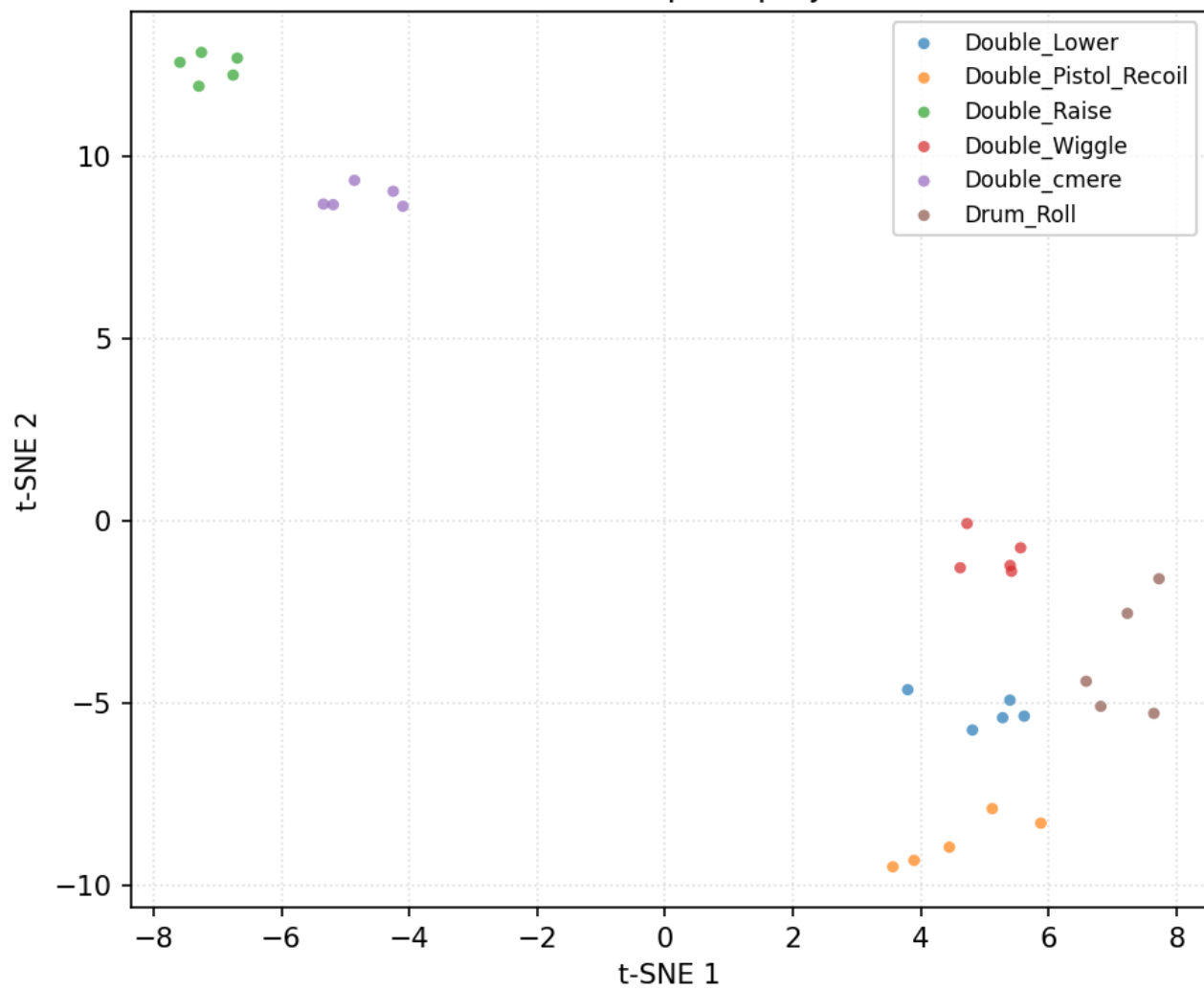
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

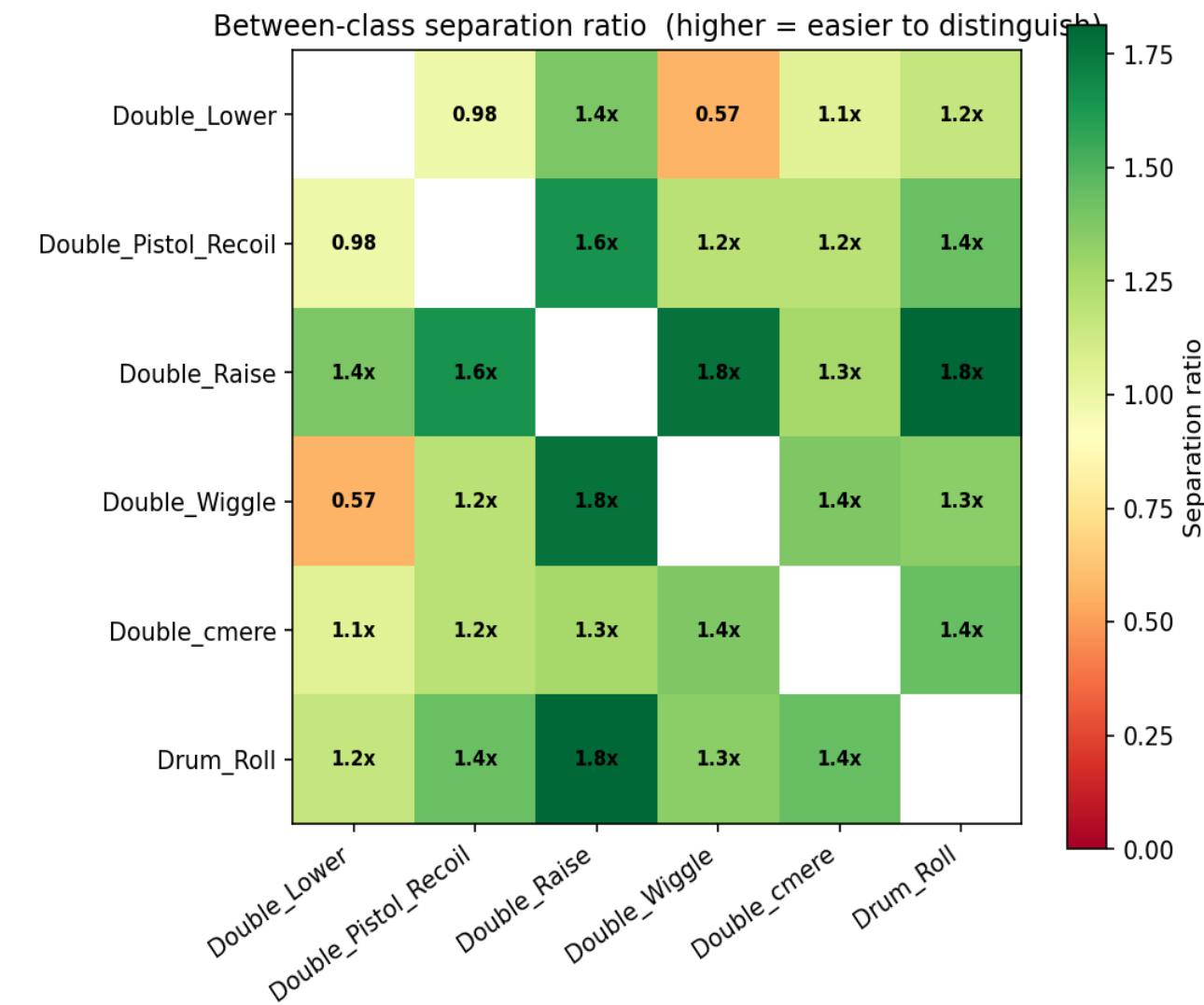
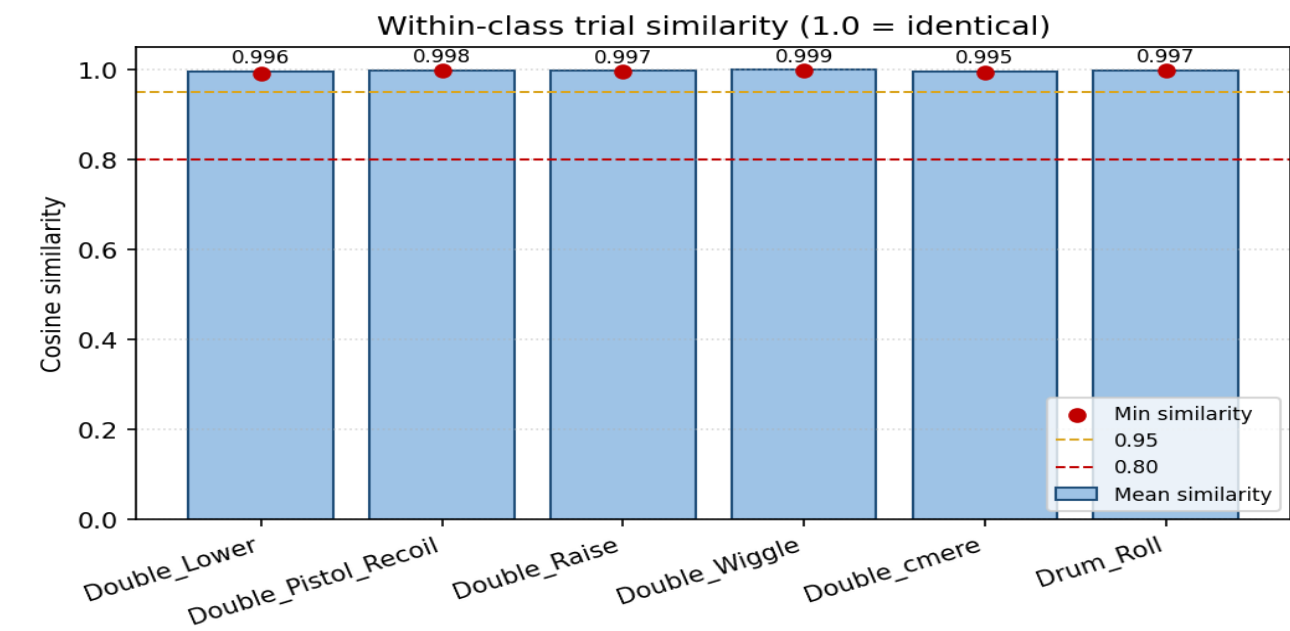


t-SNE feature space projection

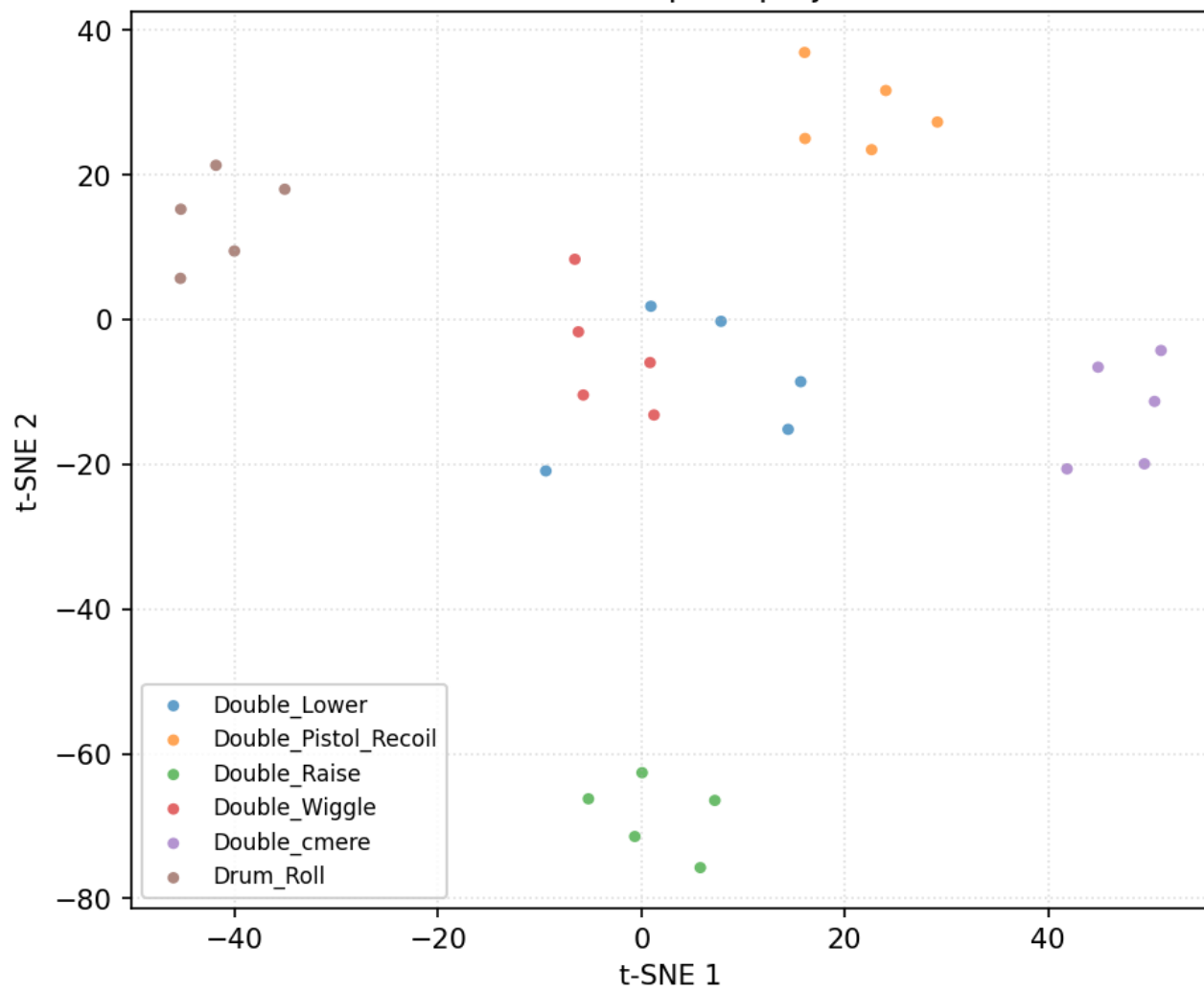


Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

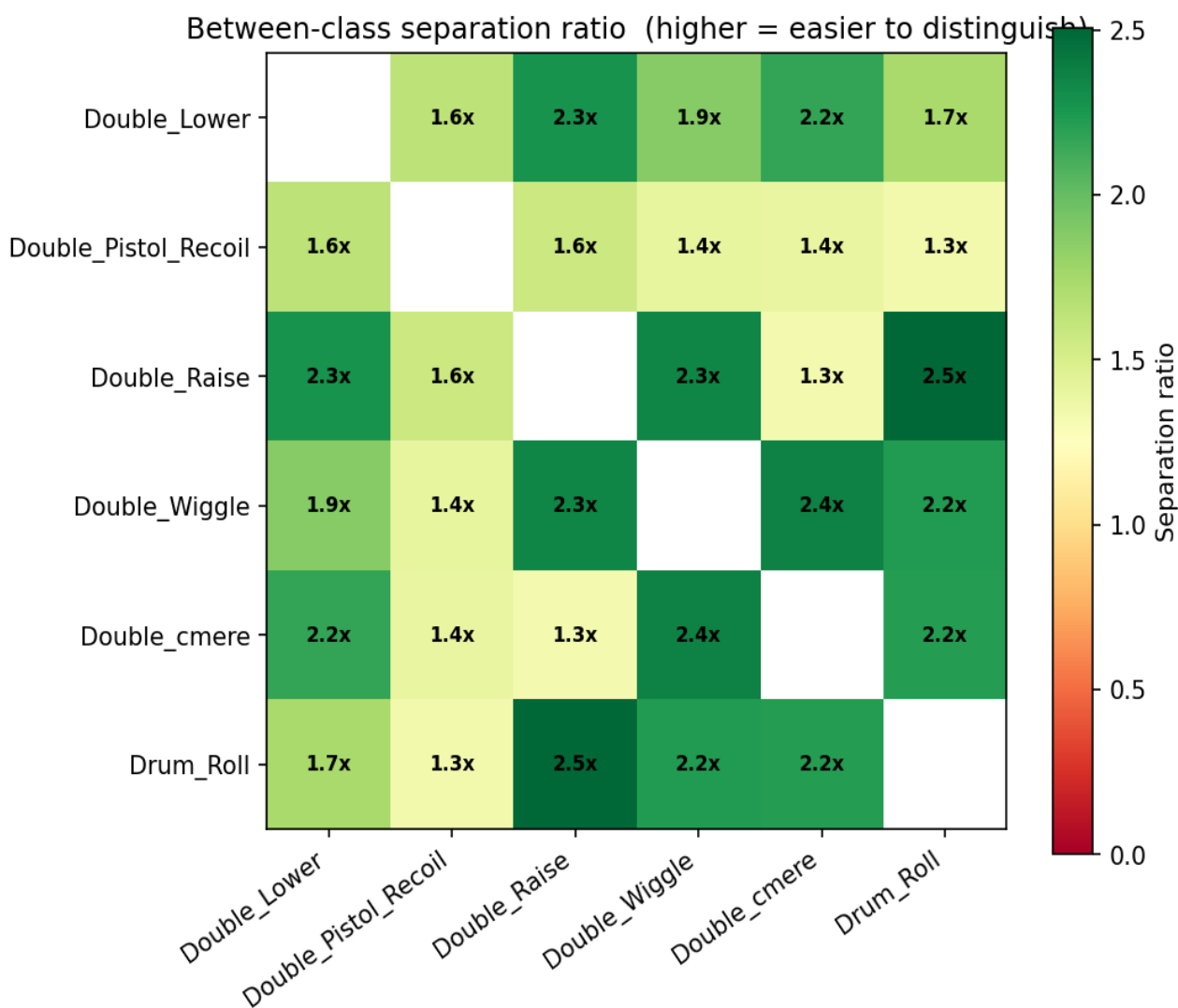
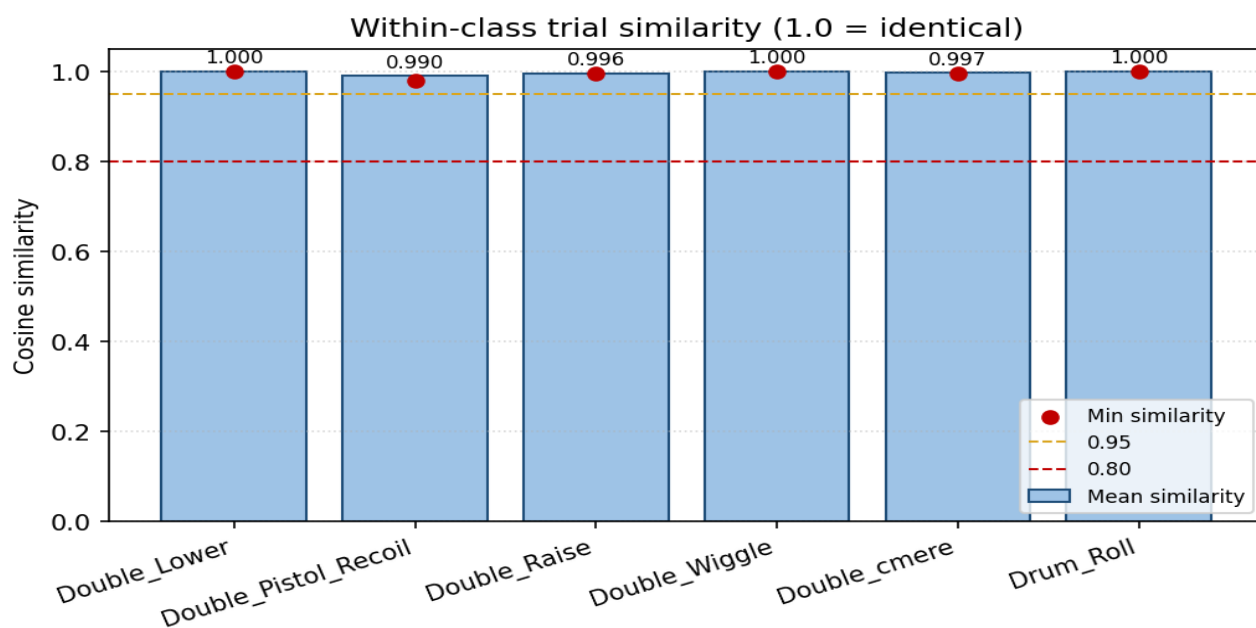


t-SNE feature space projection

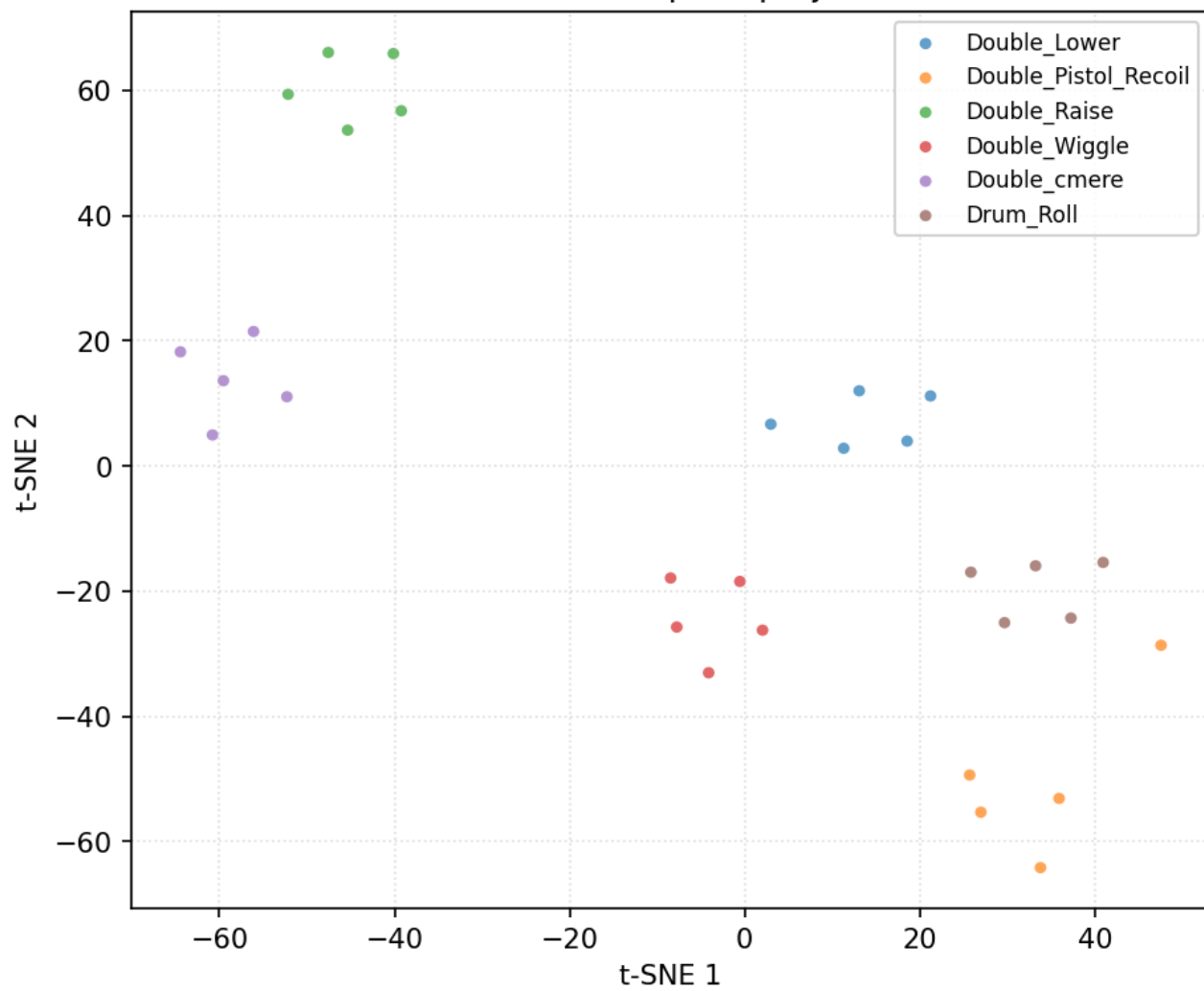


Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

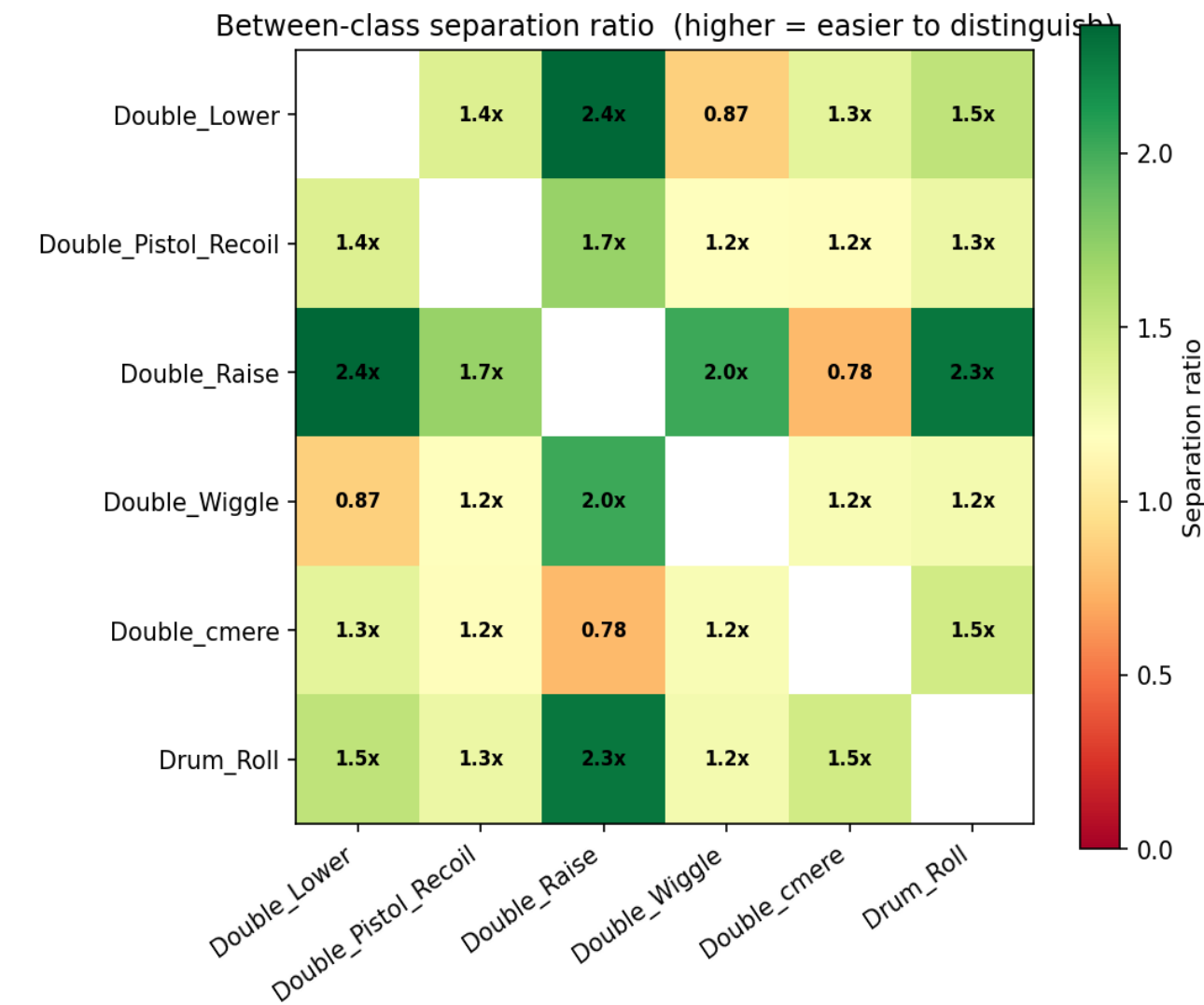
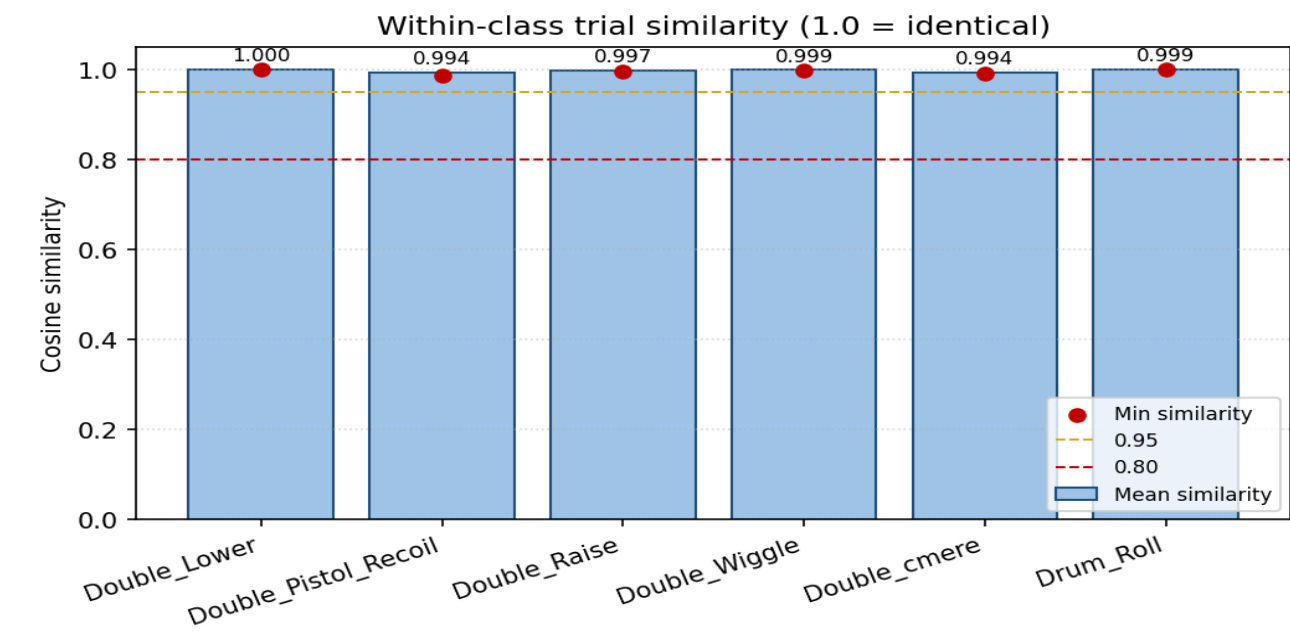


t-SNE feature space projection

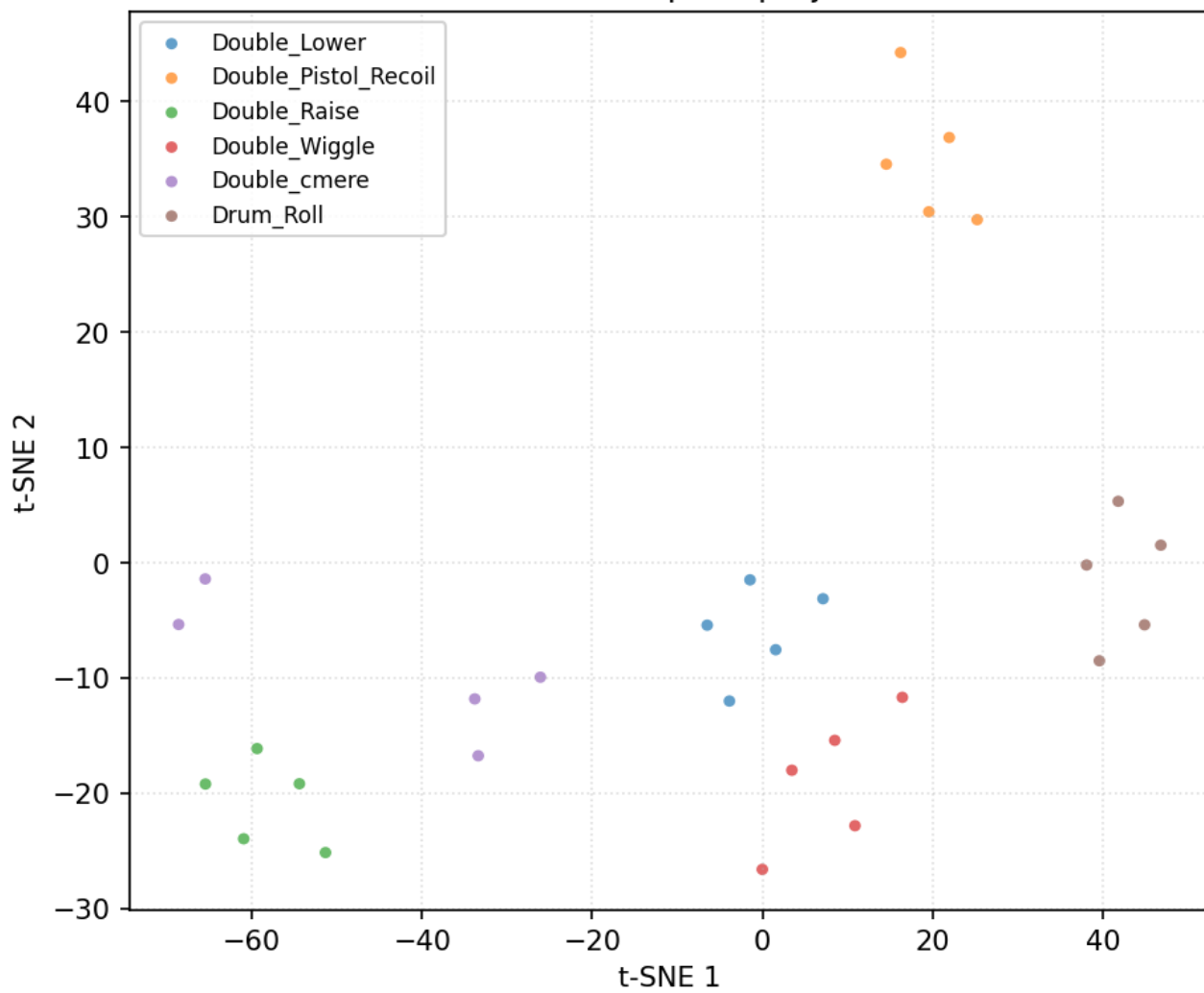


Data diagnostics — user 15_Tash

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

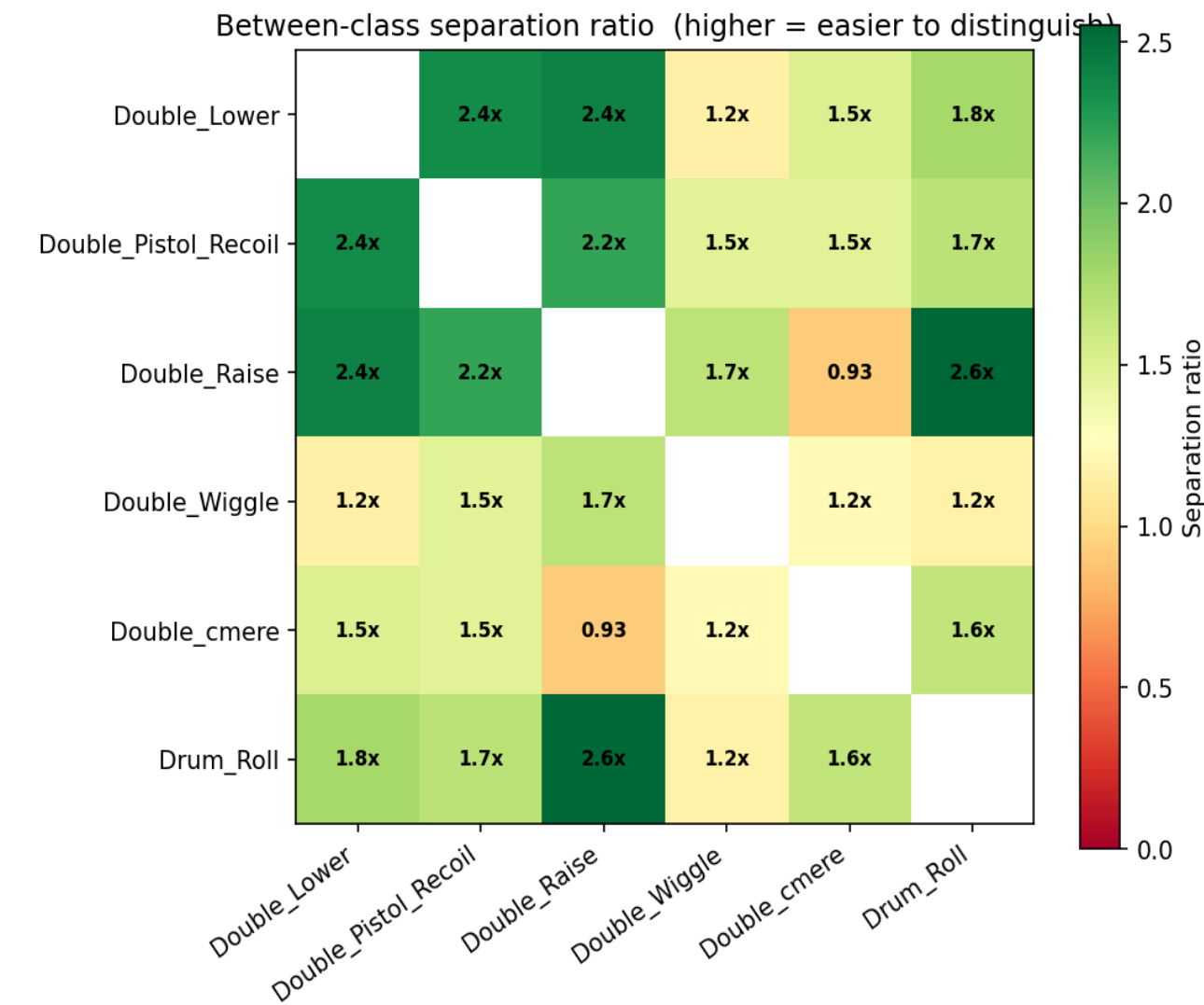
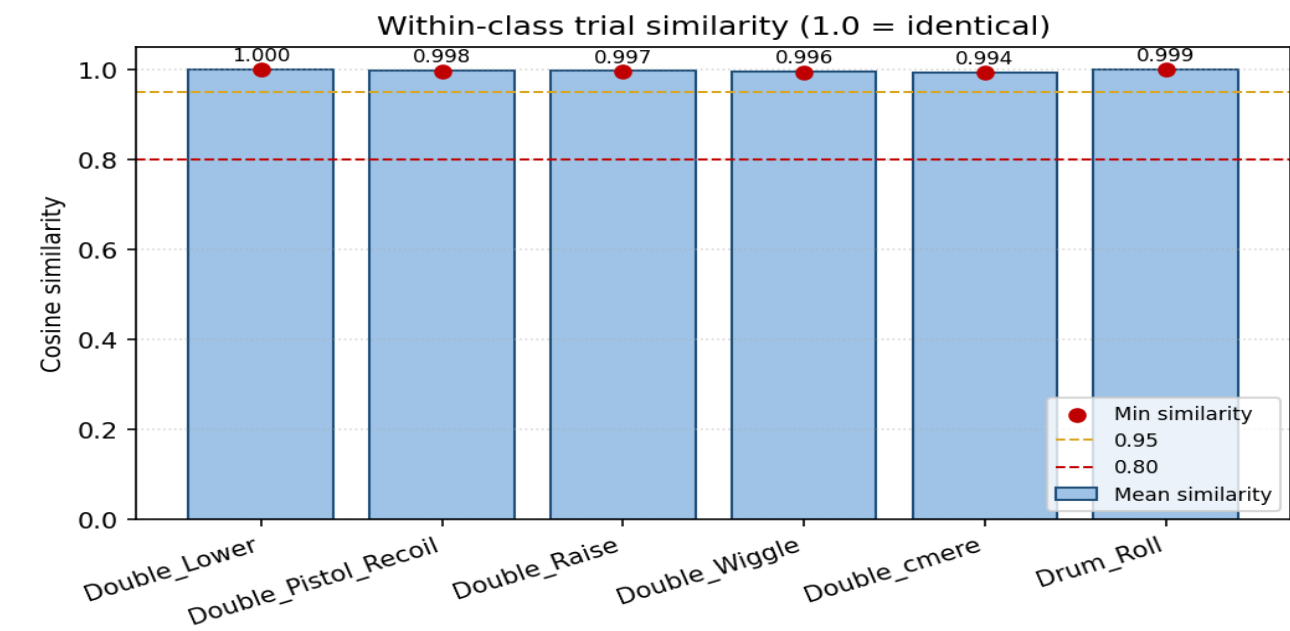


t-SNE feature space projection

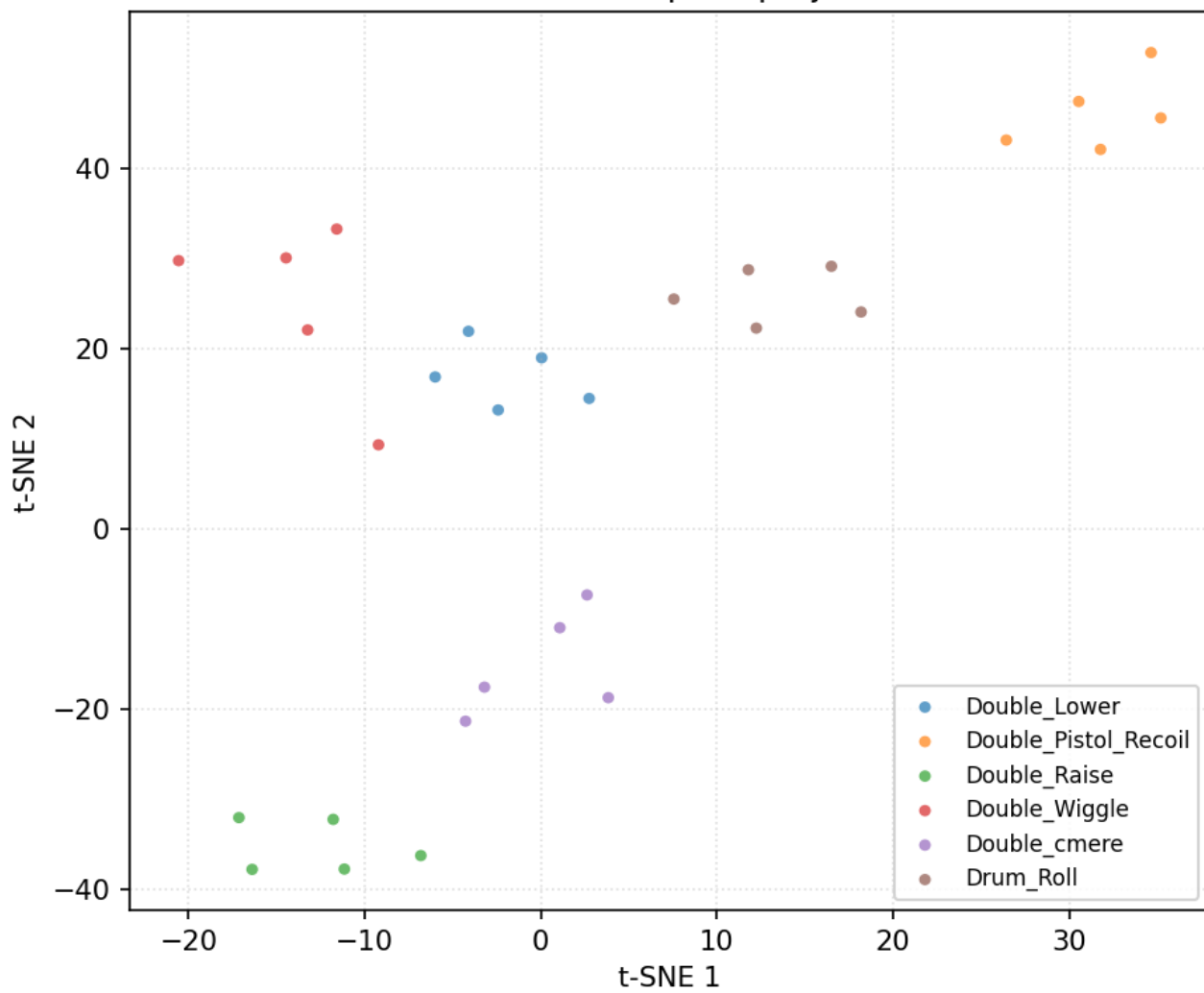


Data diagnostics — user 16_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

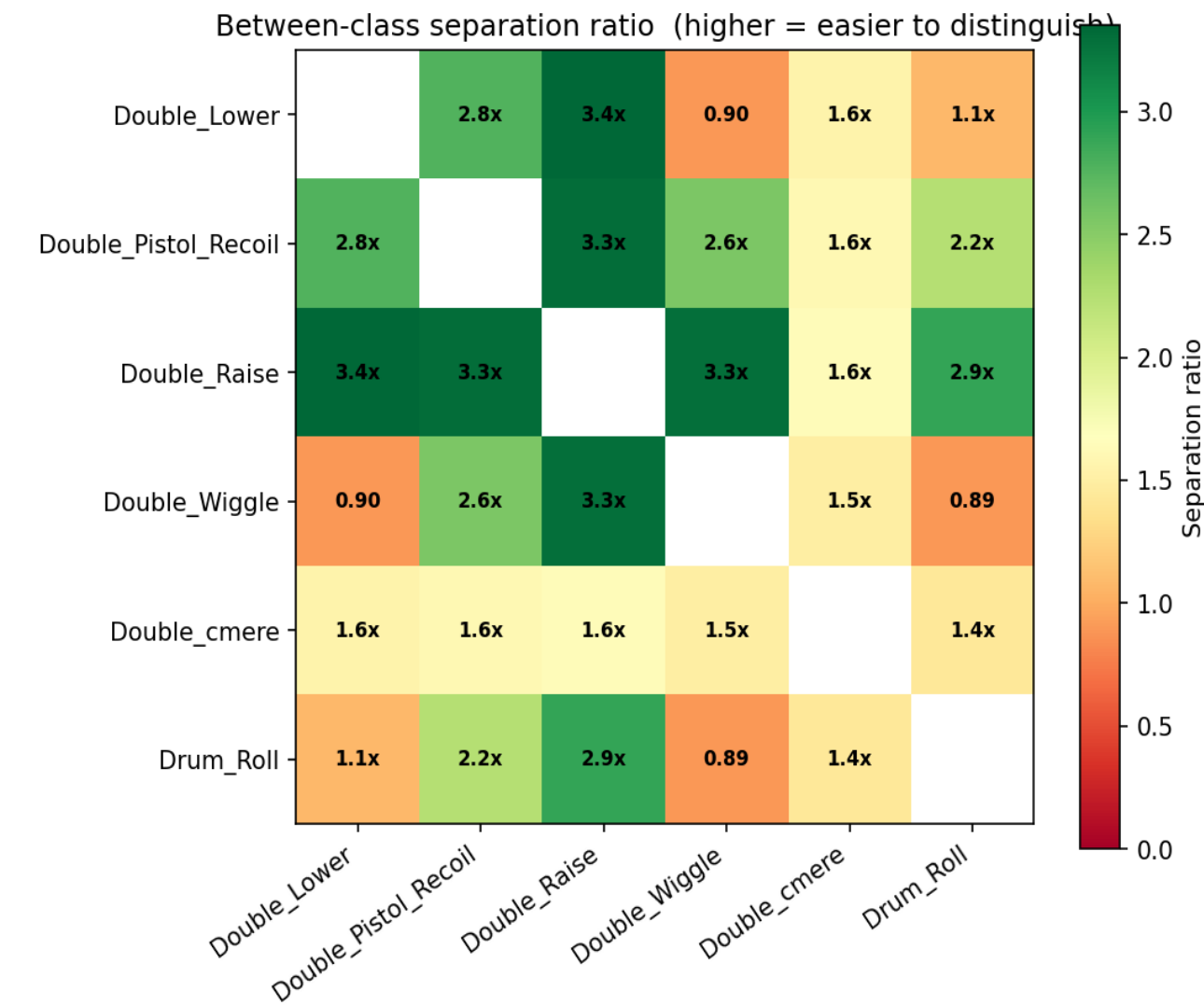
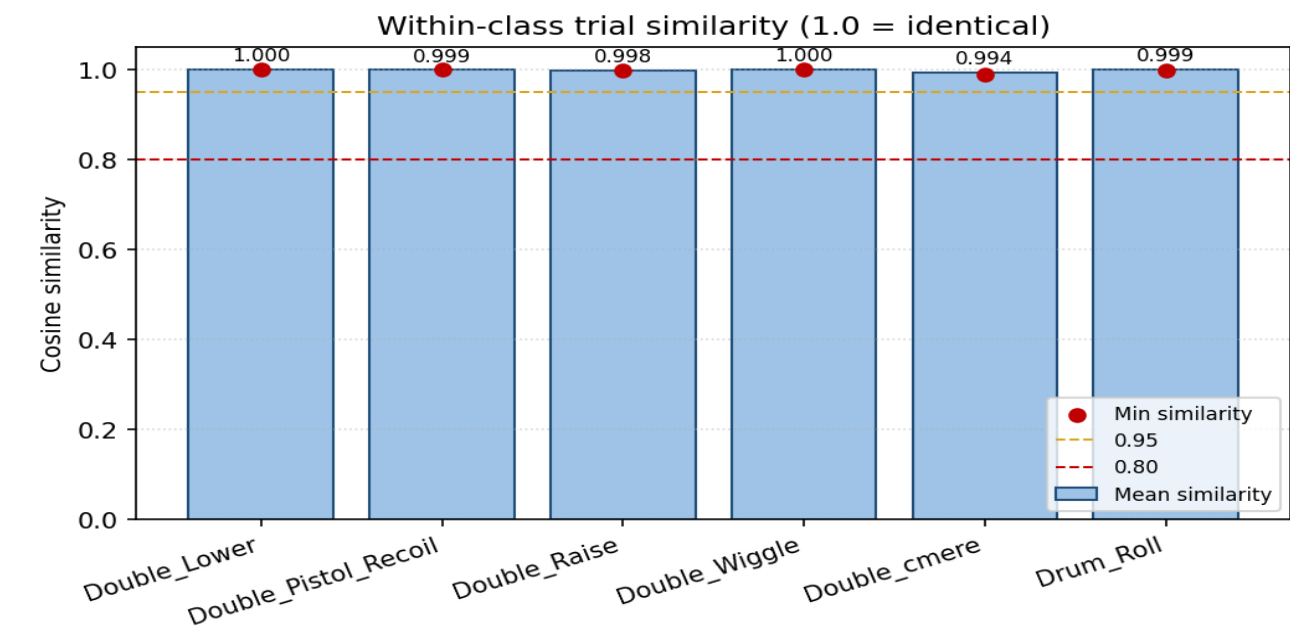


t-SNE feature space projection

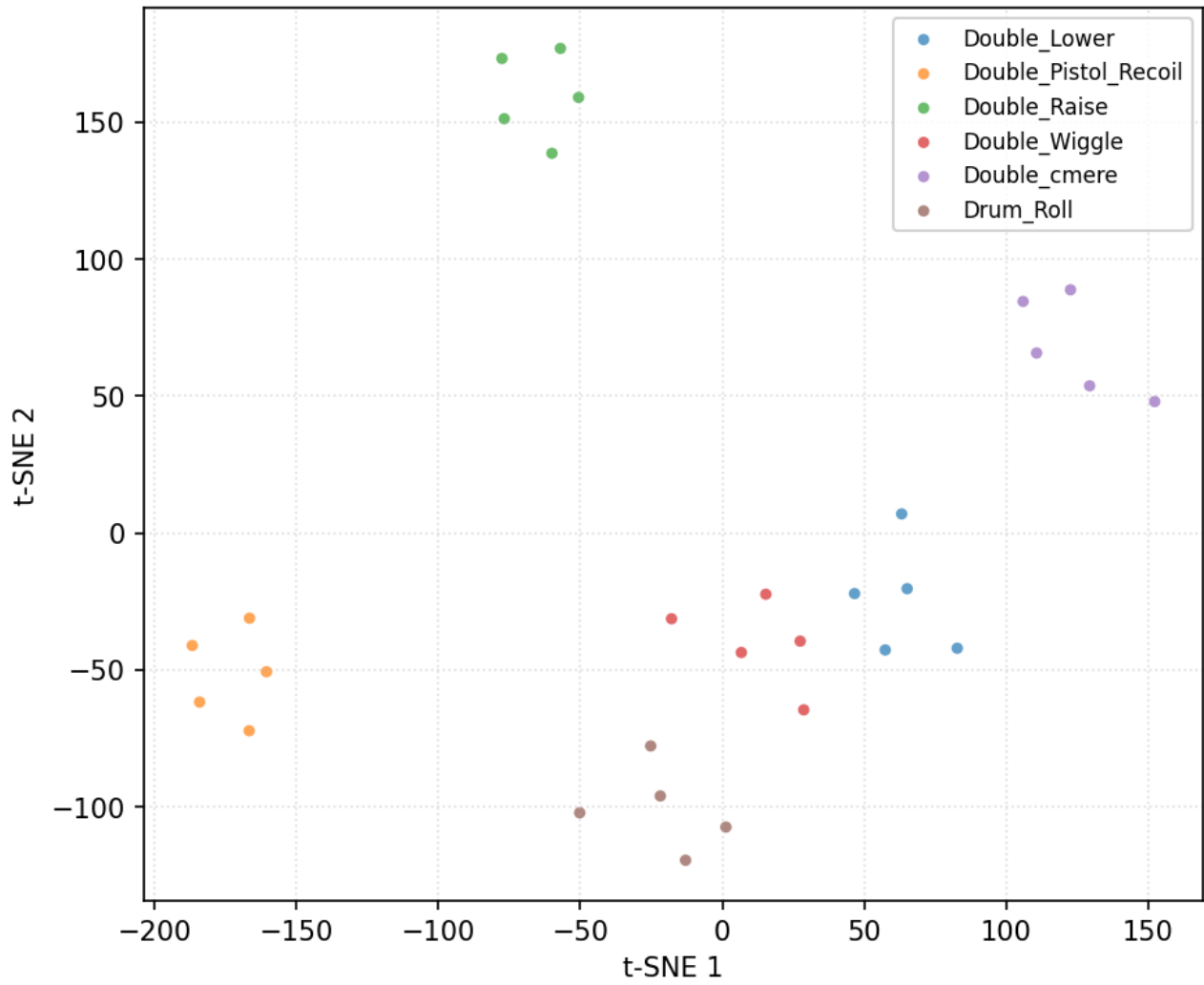


Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

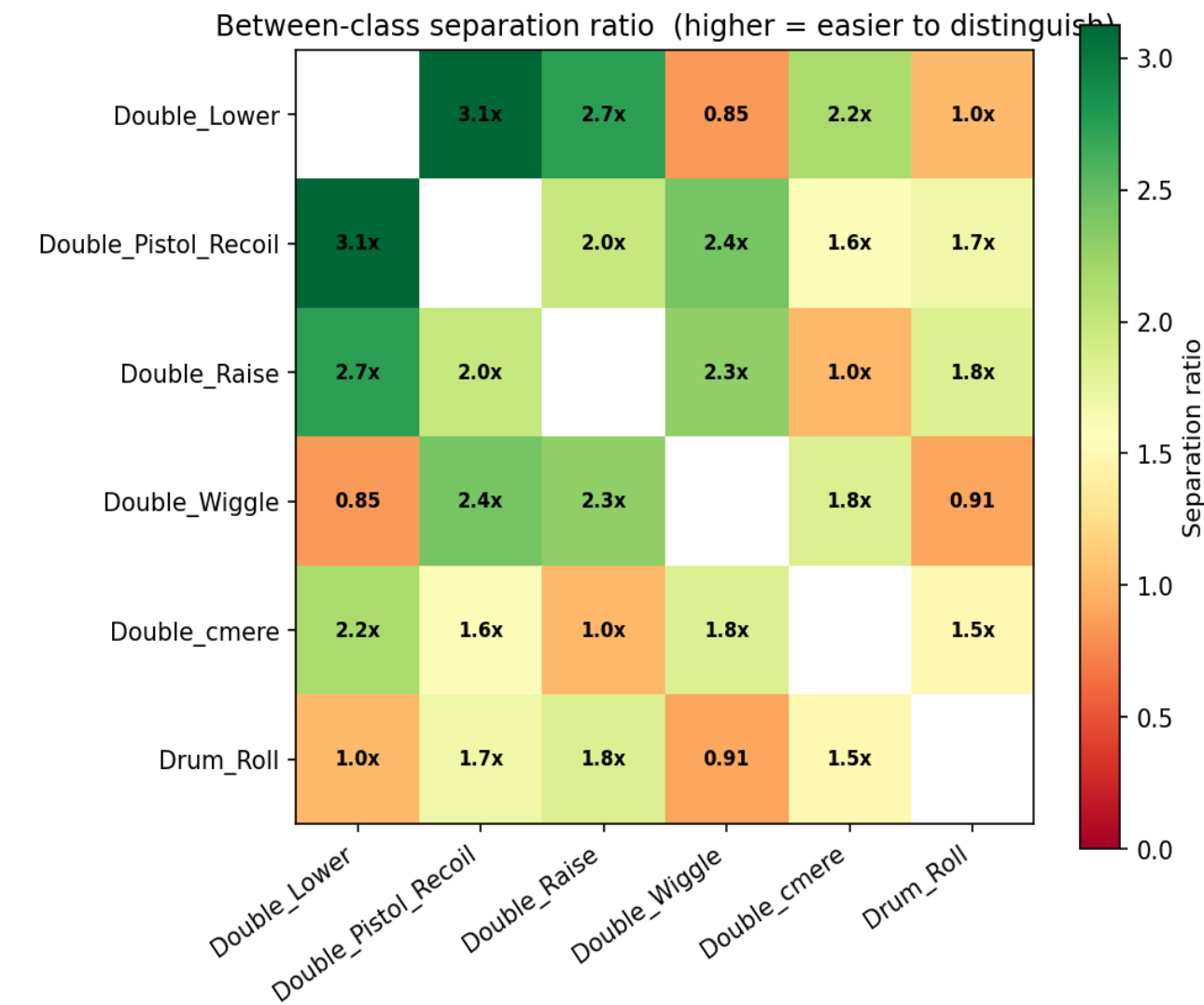
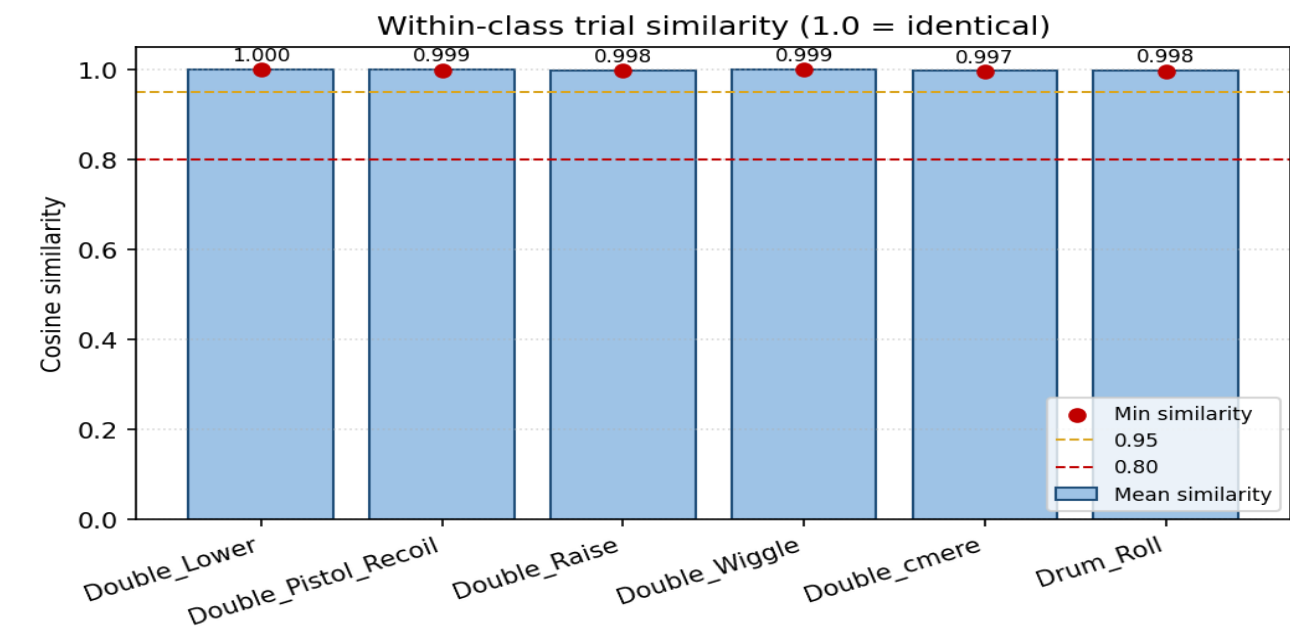


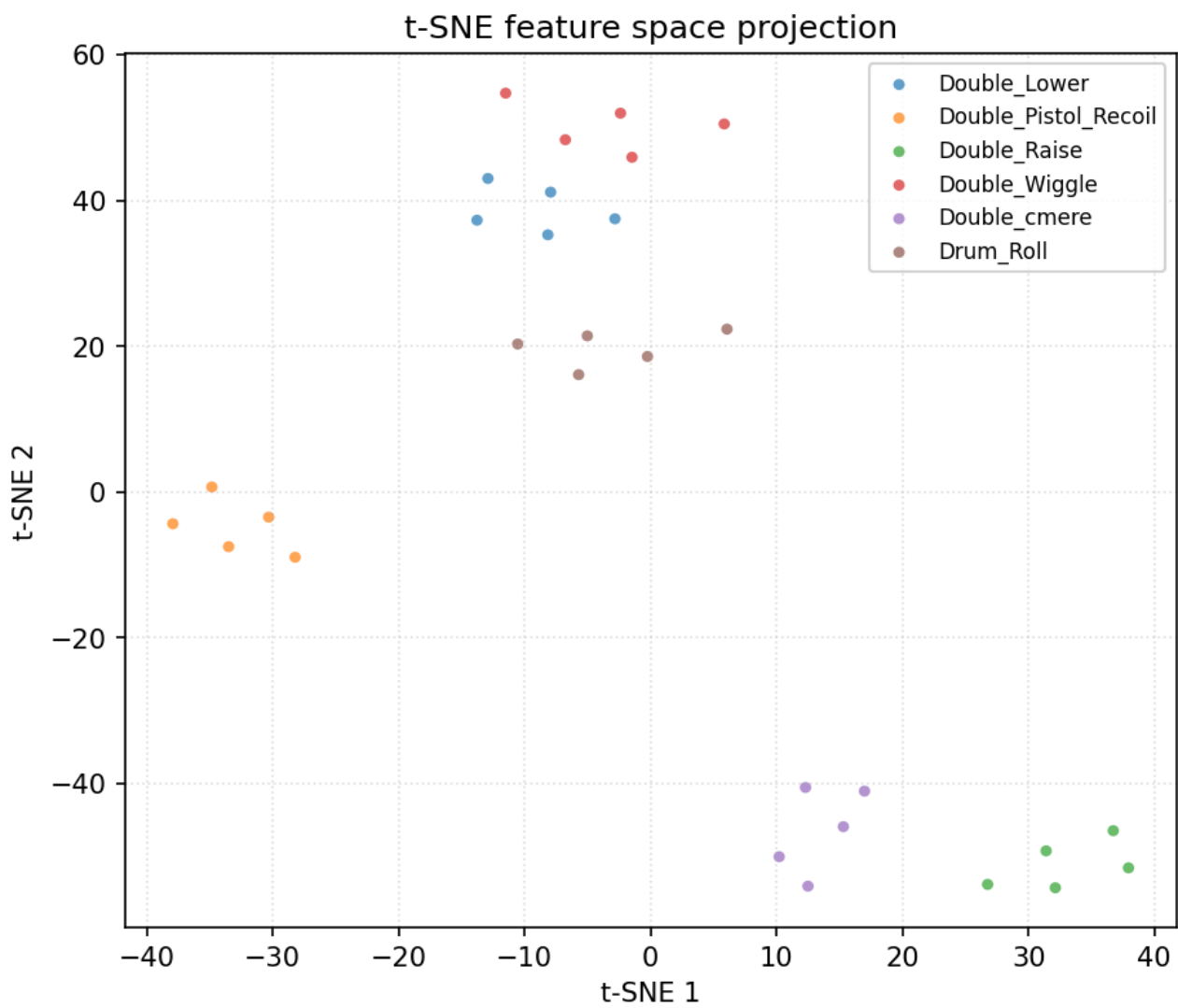
t-SNE feature space projection



Data diagnostics — user 18_Bridgette

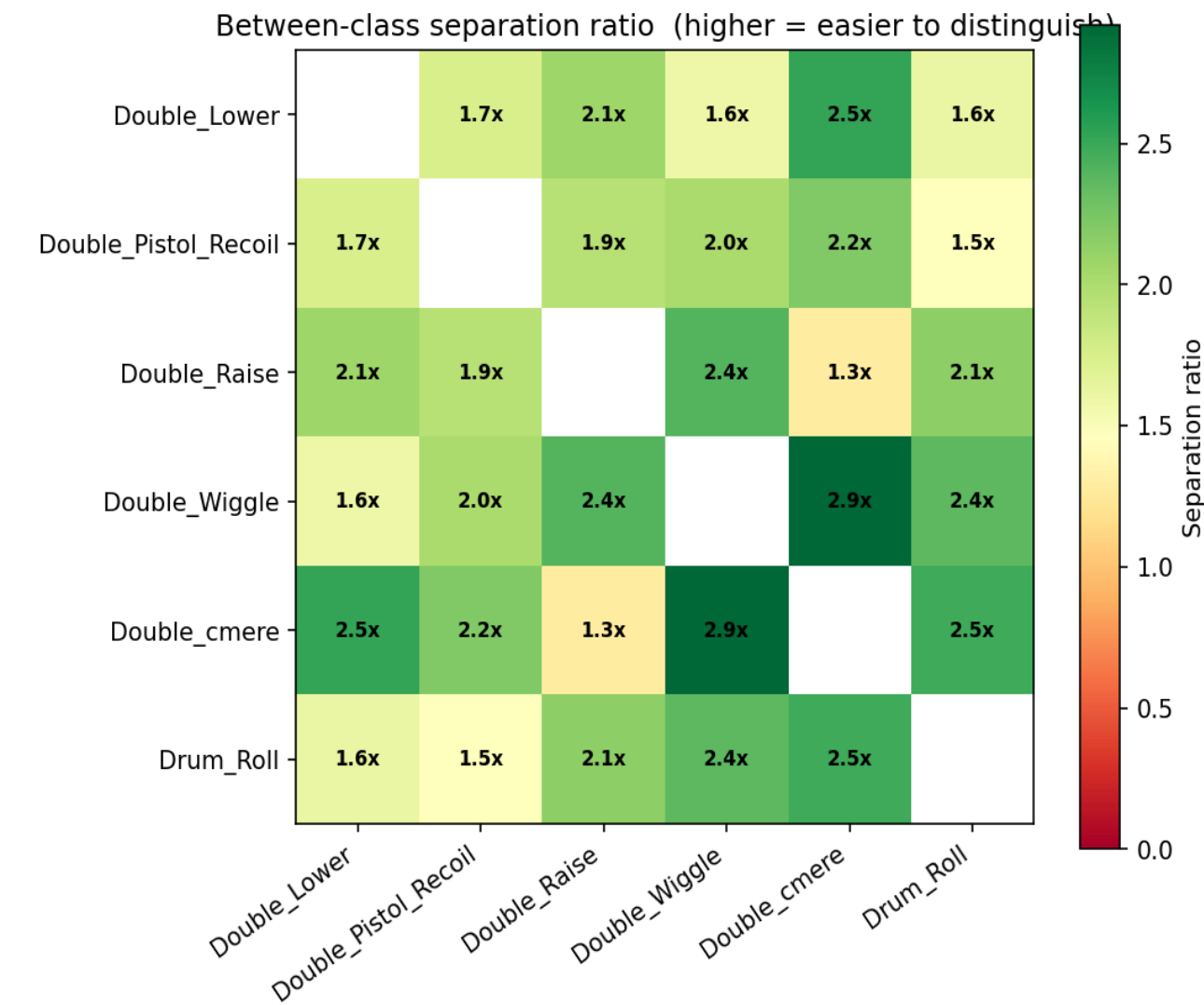
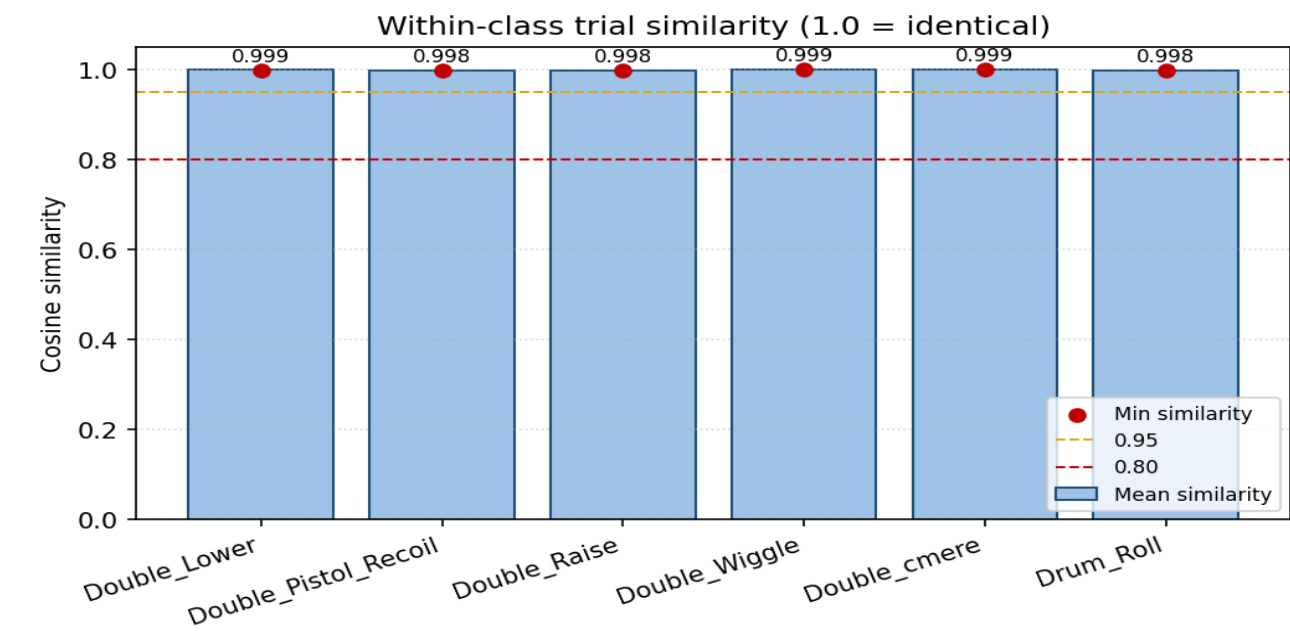
30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



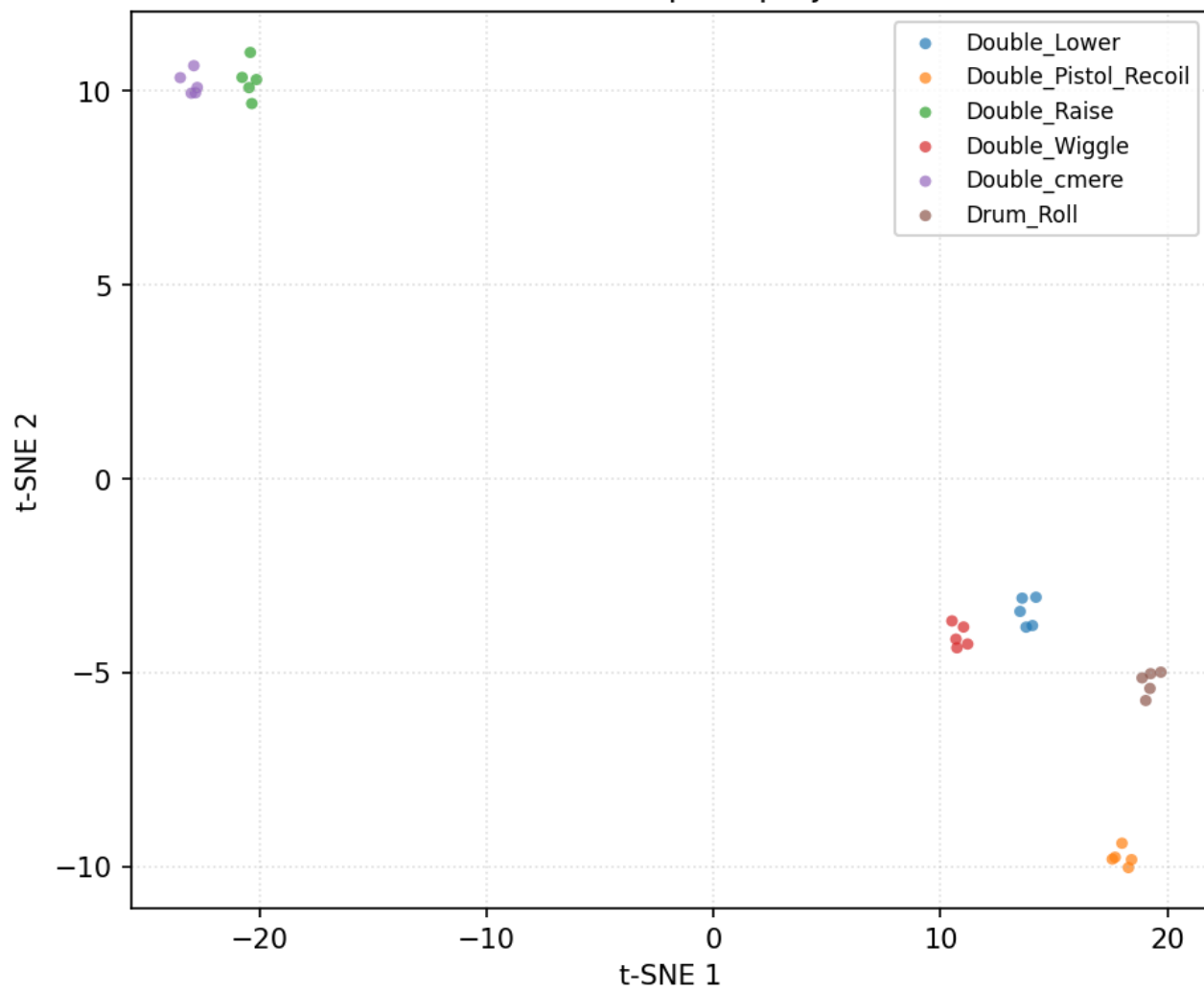


Data diagnostics — user 1_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

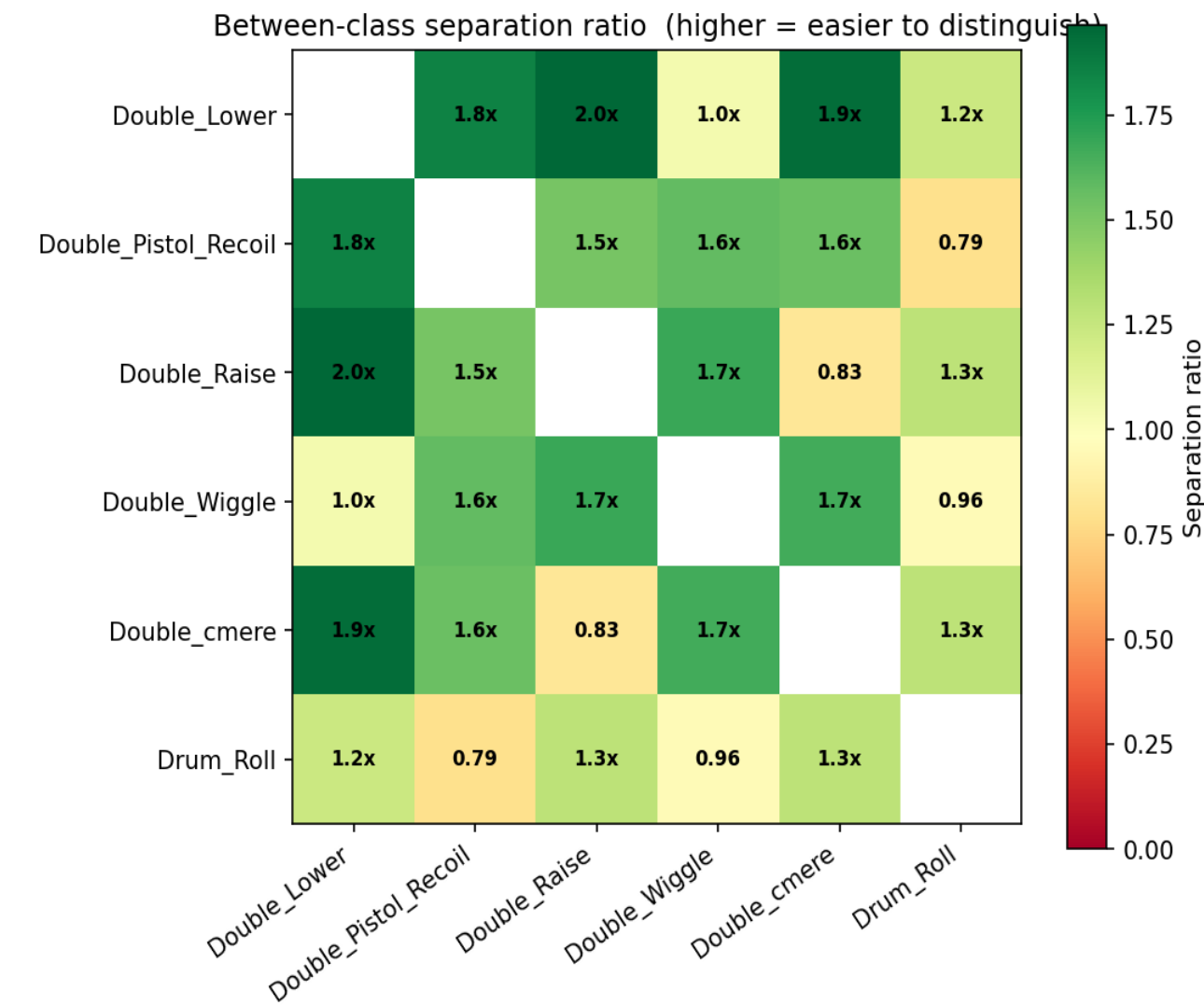
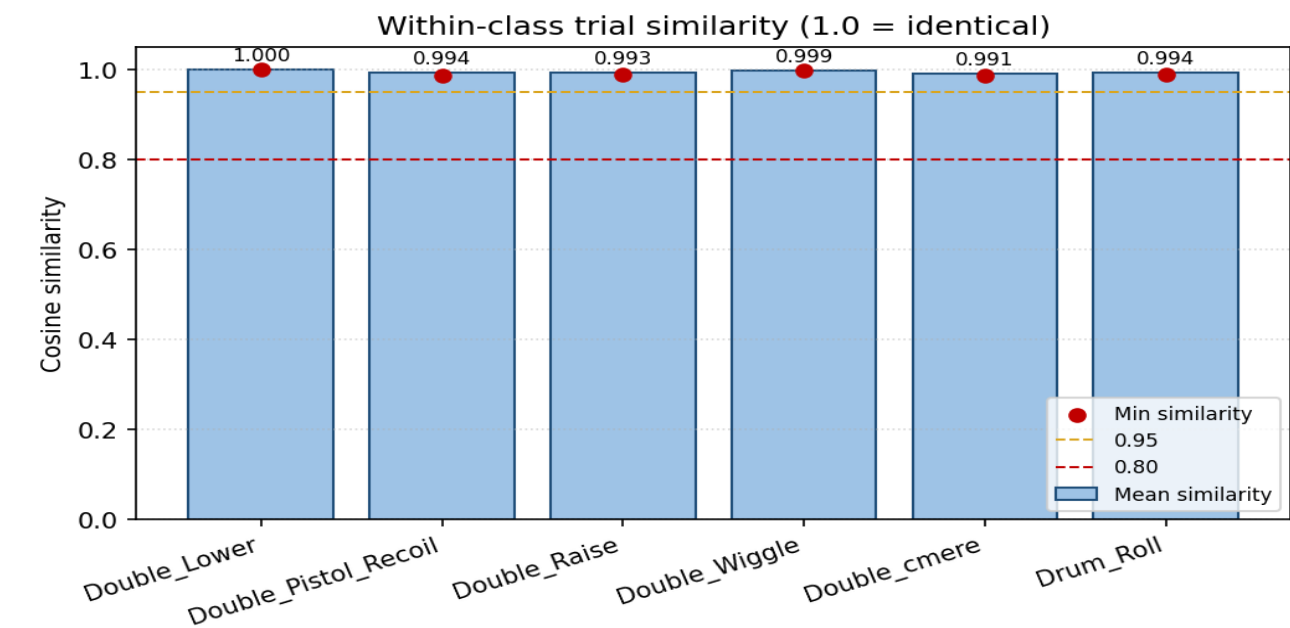


t-SNE feature space projection

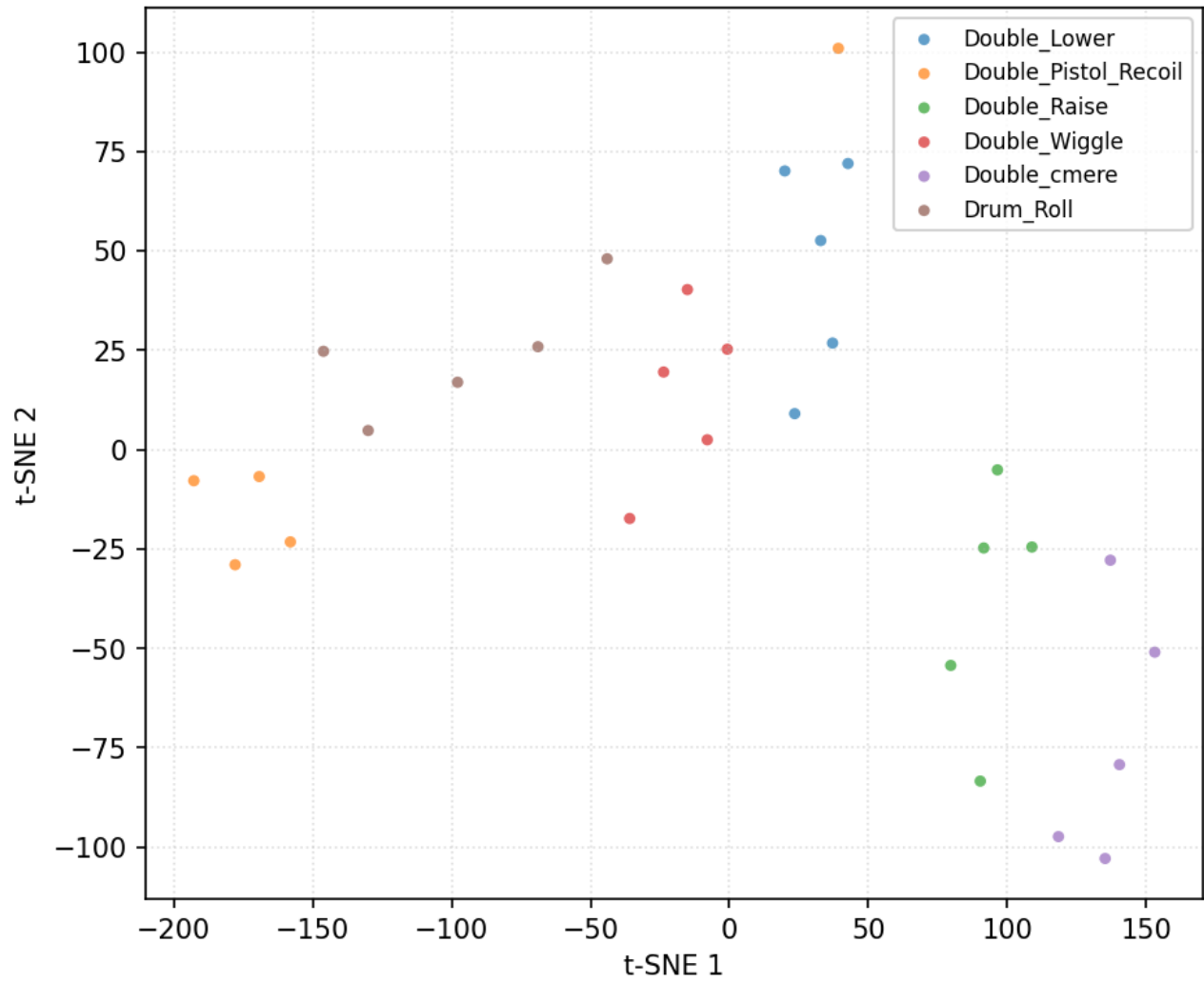


Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

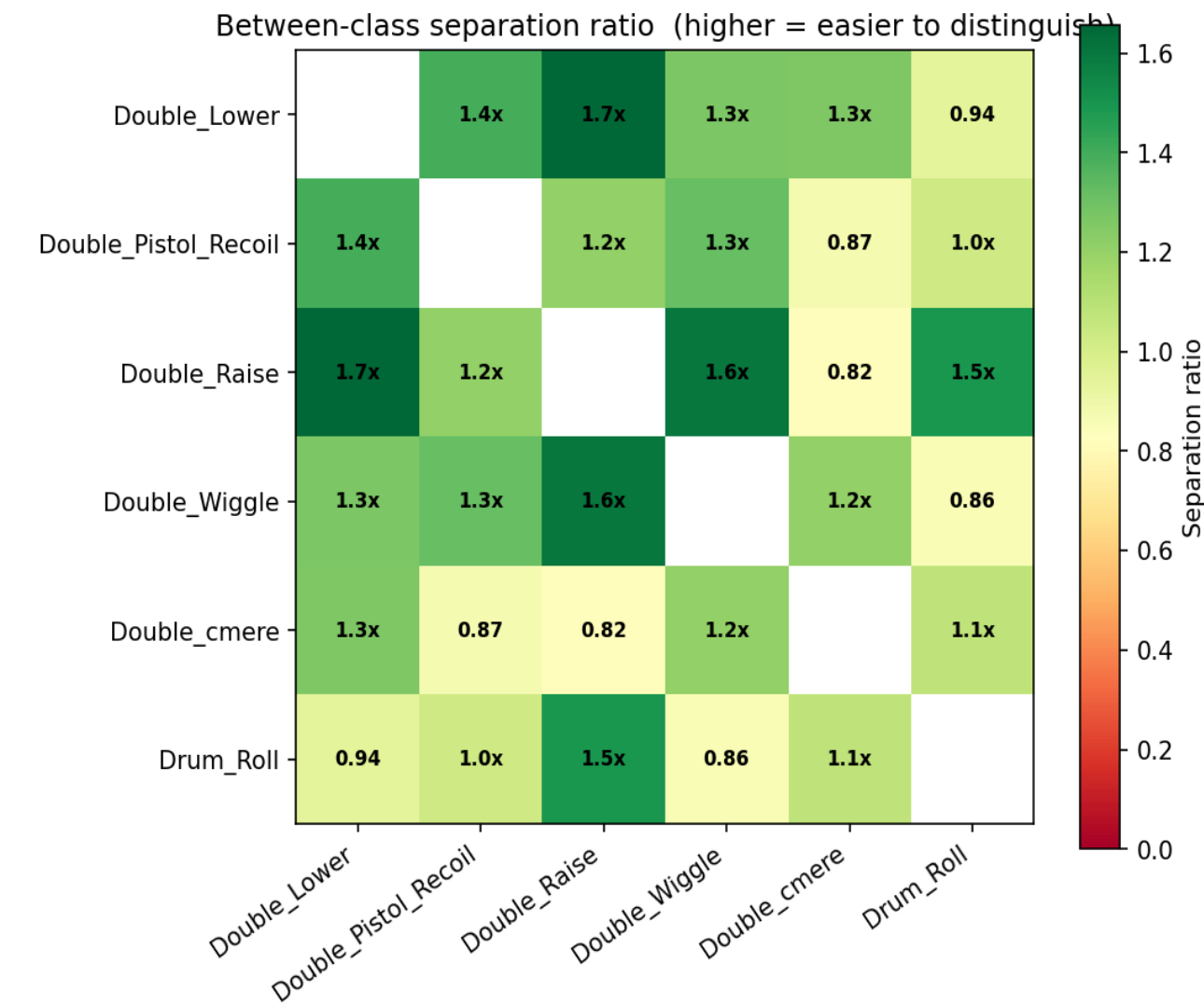
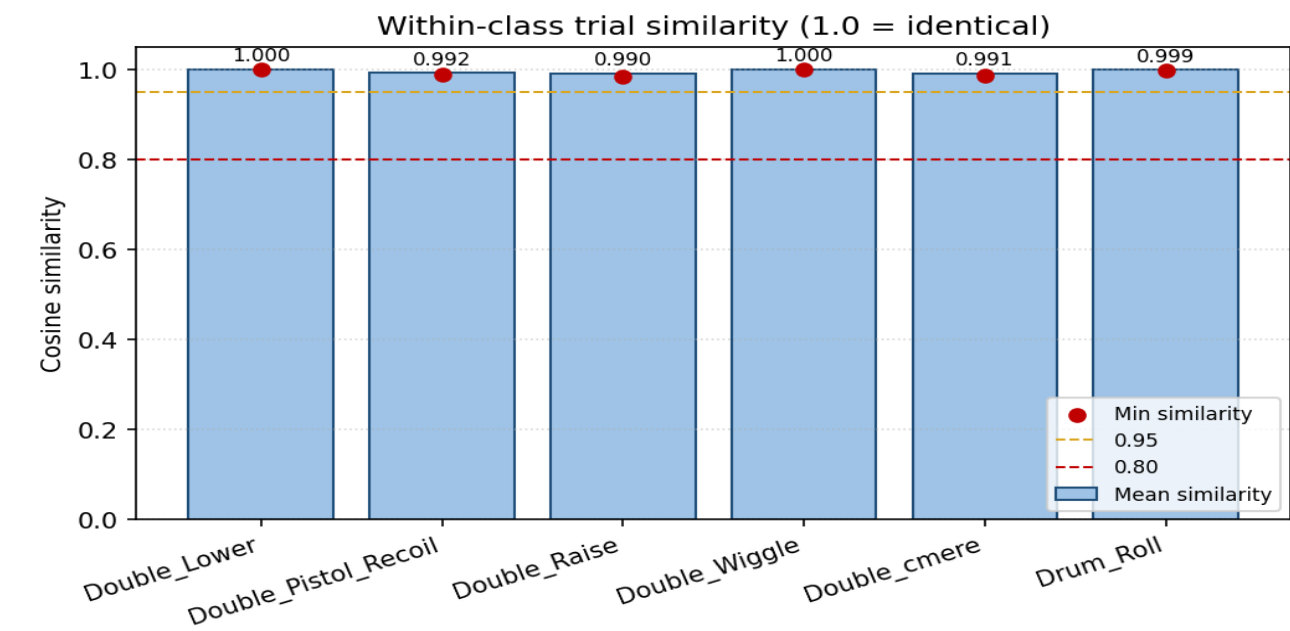


t-SNE feature space projection

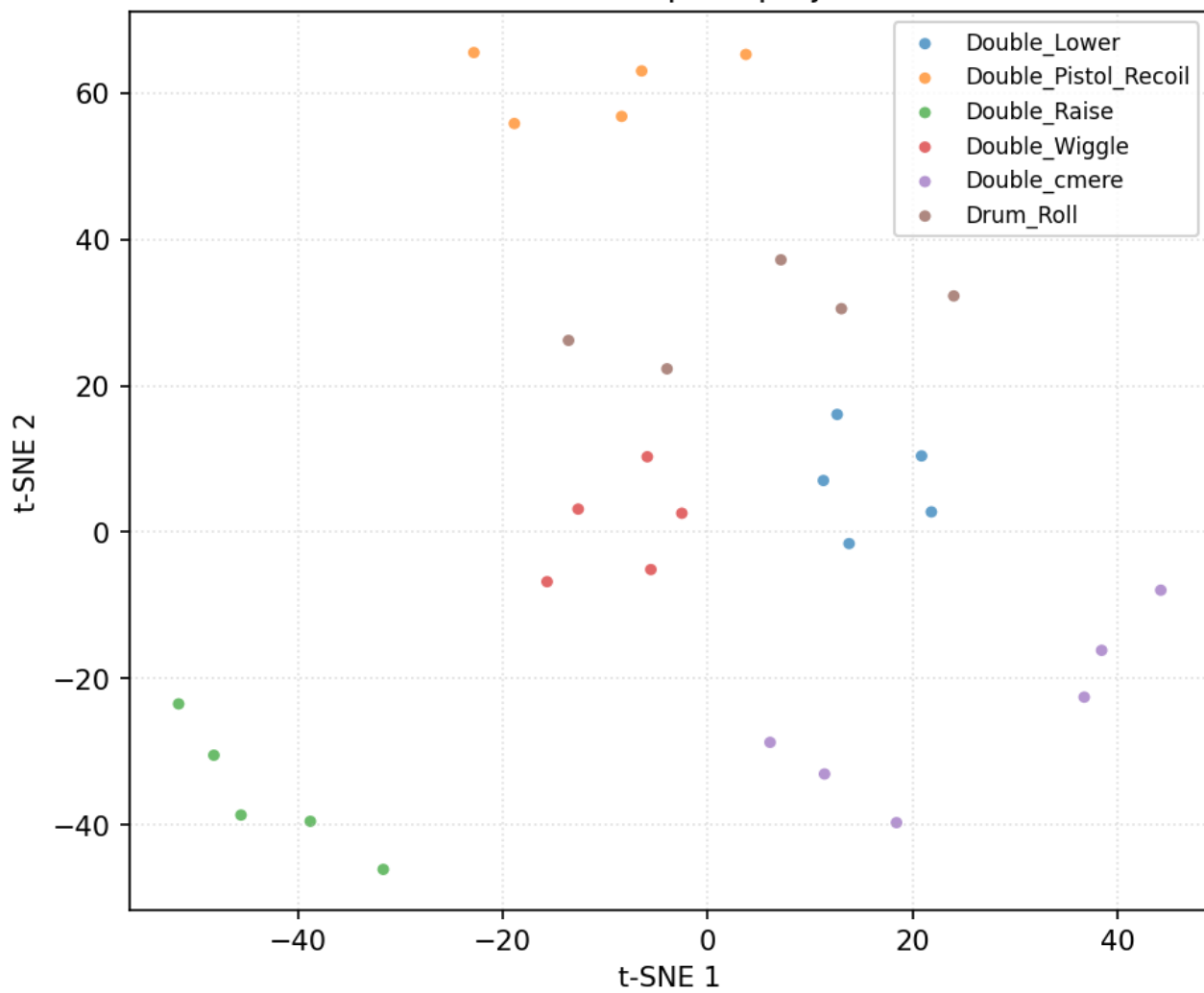


Data diagnostics — user 3_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

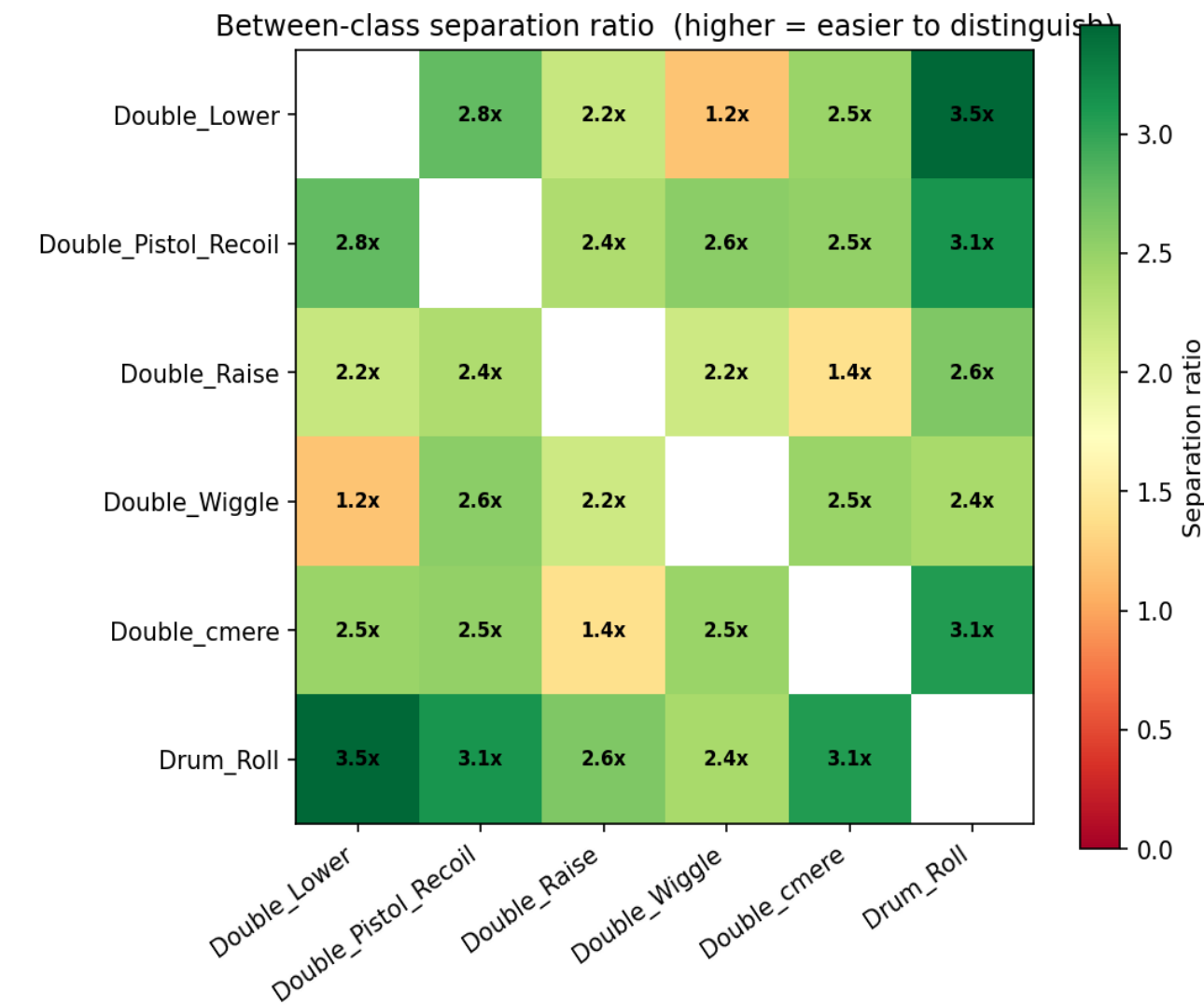
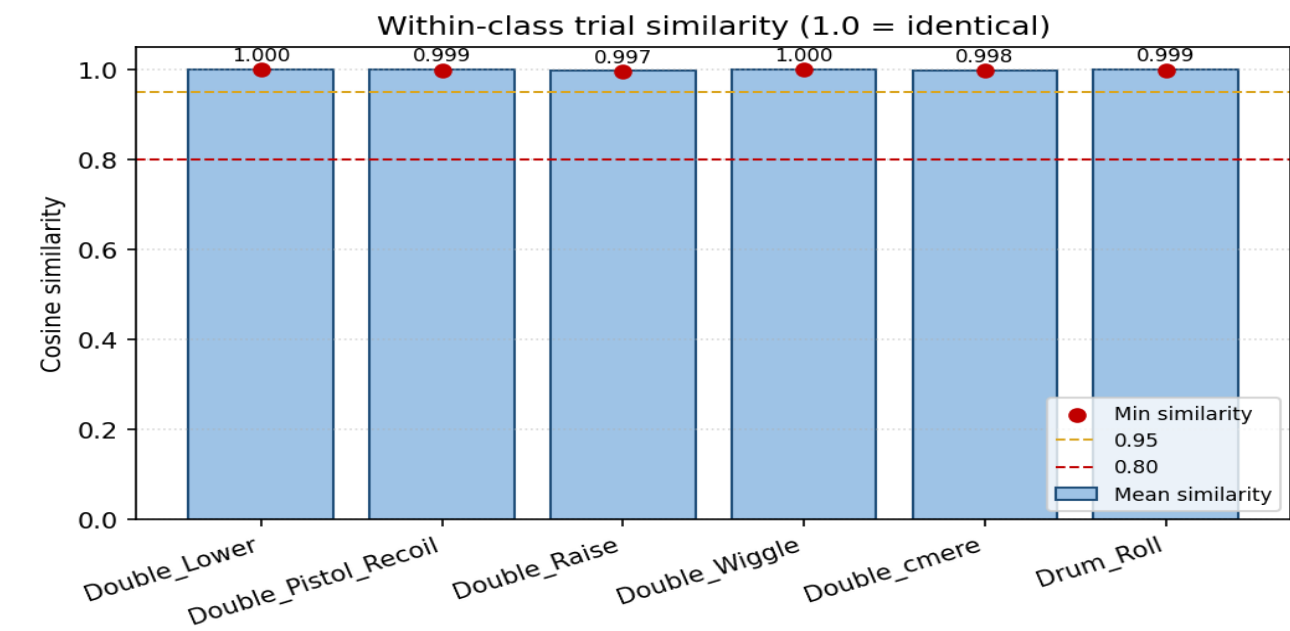


t-SNE feature space projection

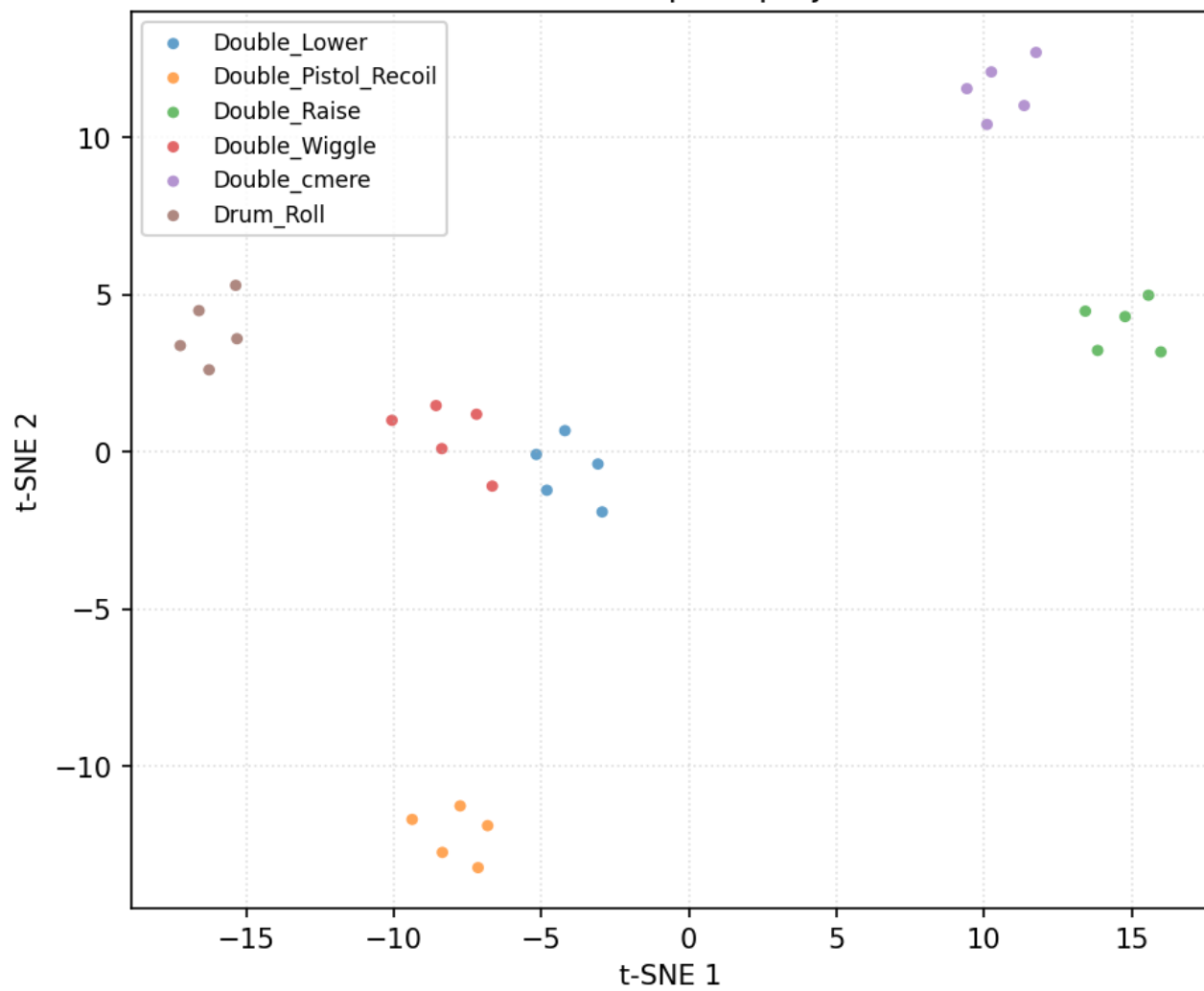


Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

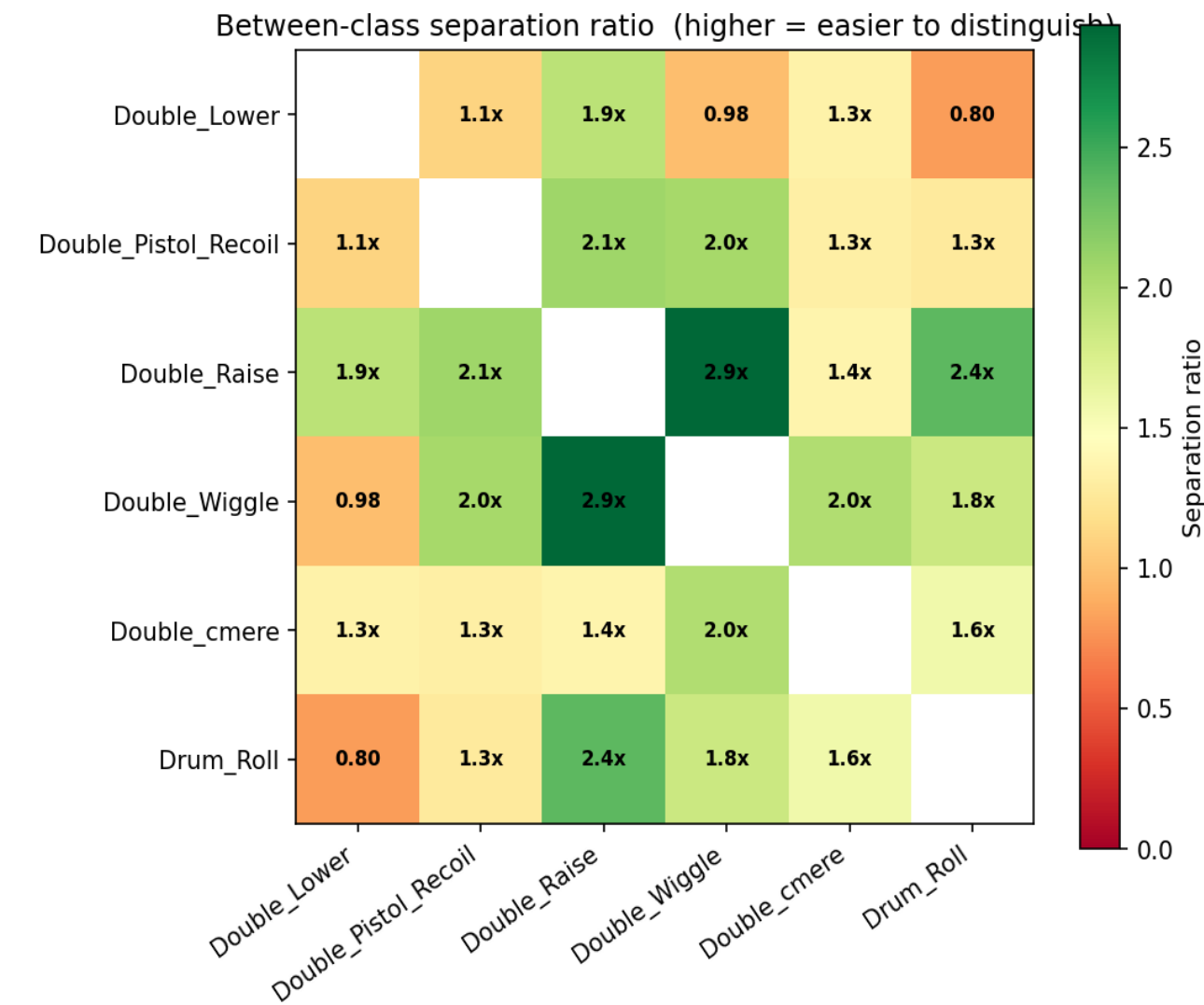
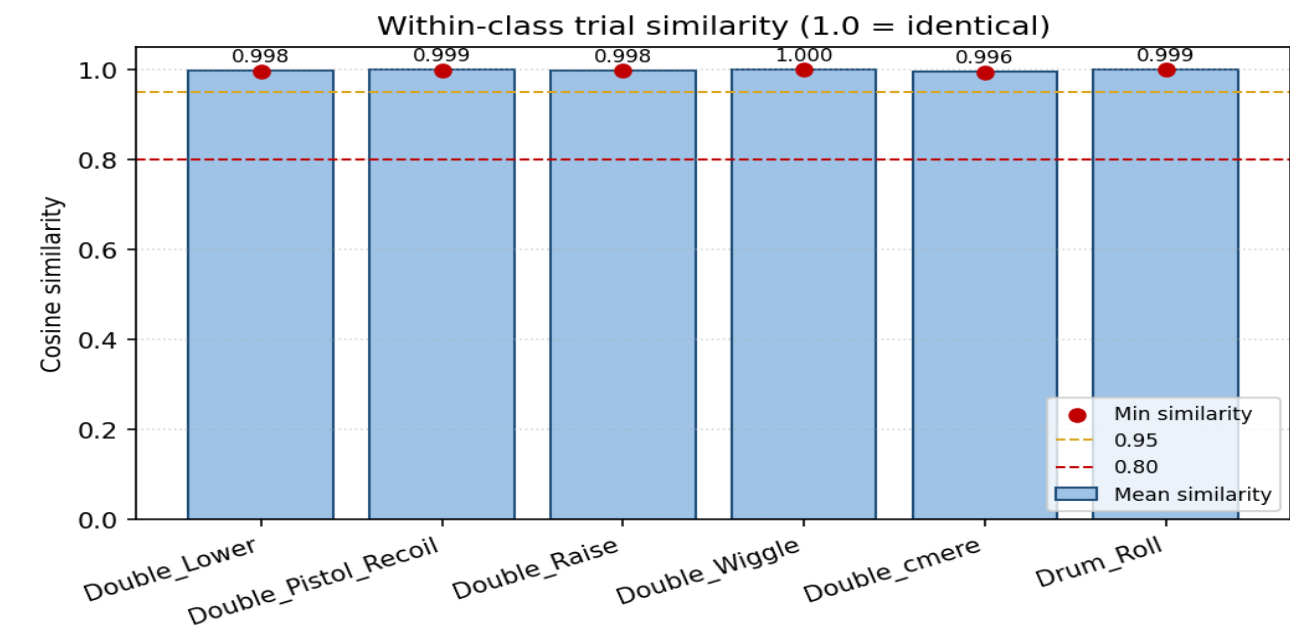


t-SNE feature space projection

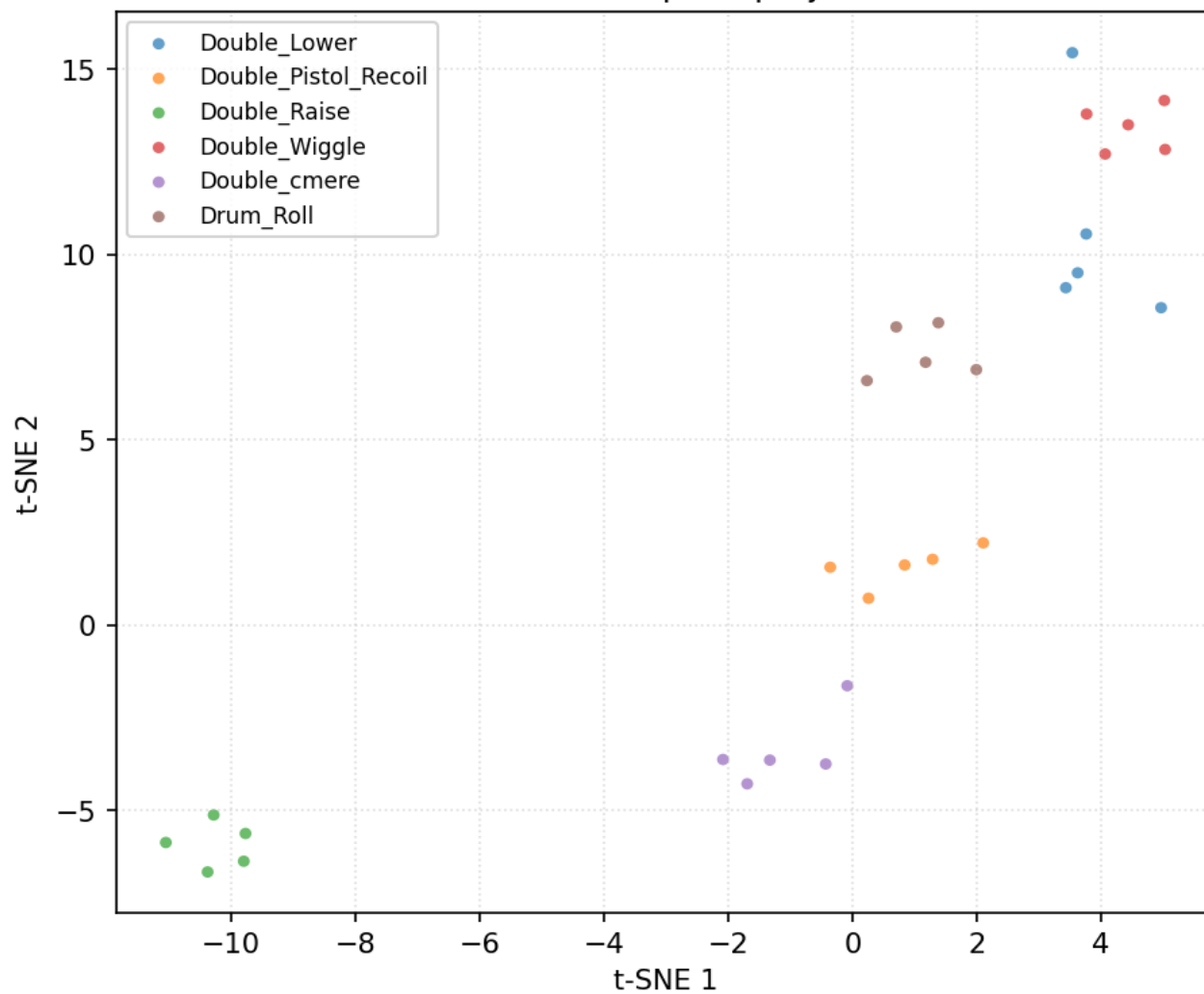


Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

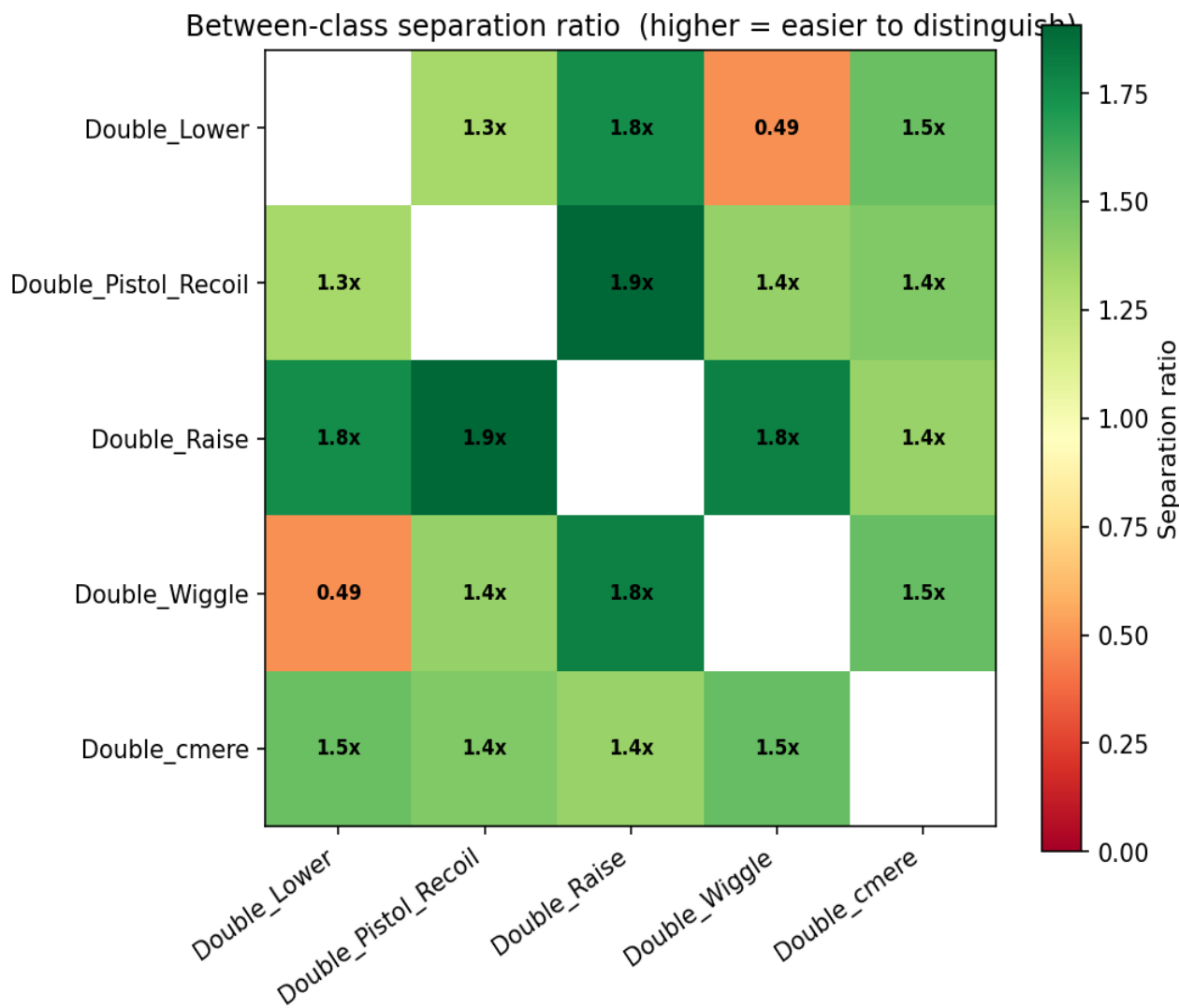
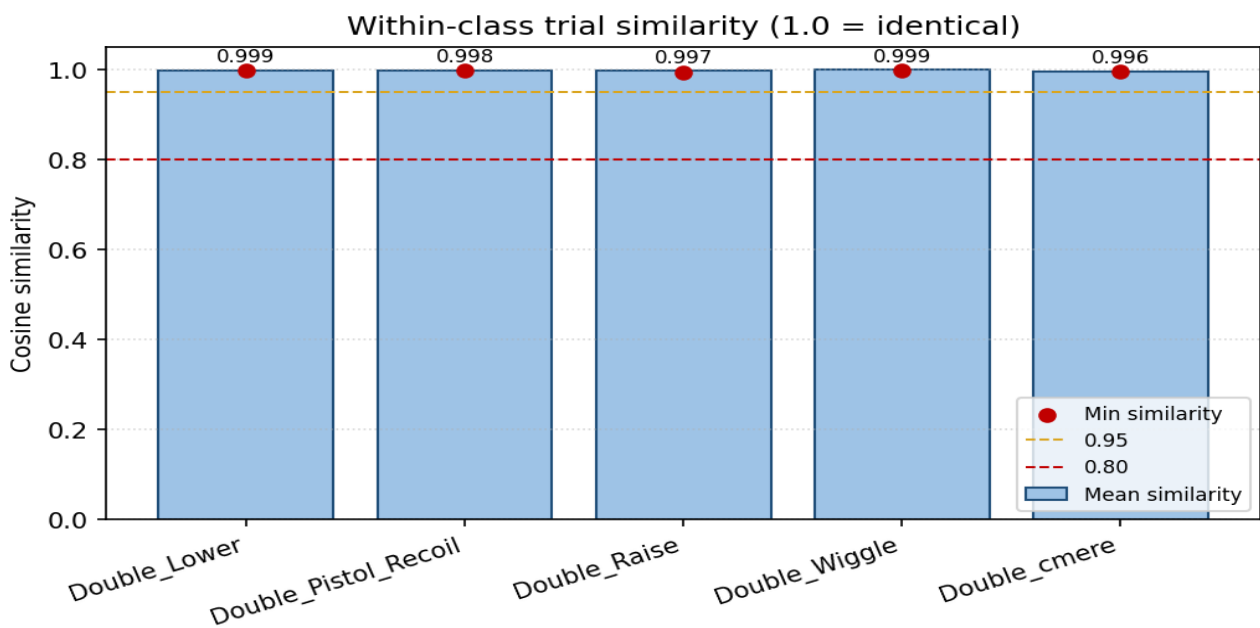


t-SNE feature space projection

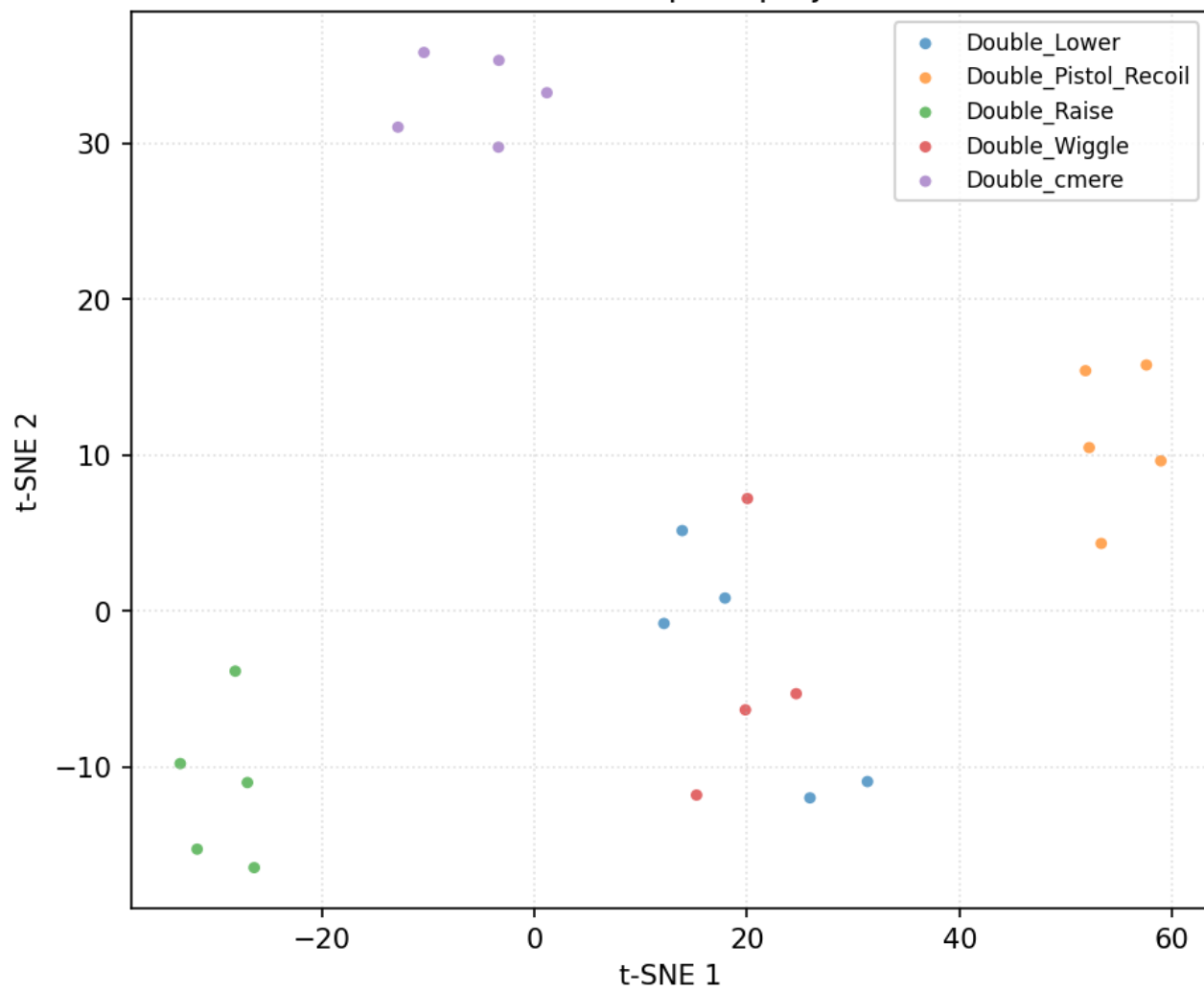


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
LOUO mean ± std	0.9942 ± 0.0136
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

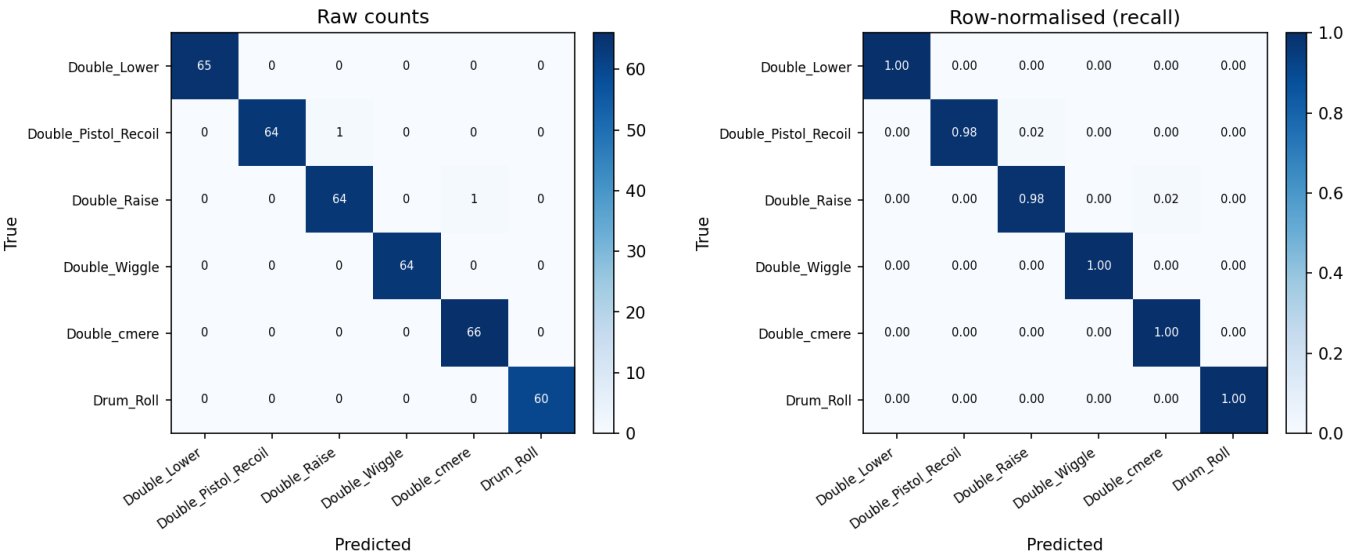
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	1.0000
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	1.0000
9	2_Alex	3_Anghad	324	30	1.0000
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	1.0000
13	8_Jestin	11_Mansh	331	24	0.9583

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	65
Double_Pistol_Recoil	1.000	0.985	0.992	65
Double_Raise	0.985	0.985	0.985	65
Double_Wiggle	1.000	1.000	1.000	64
Double_cmere	0.985	1.000	0.992	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.995	0.995	0.995	385
weighted avg	0.995	0.995	0.995	385

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
LOUO mean ± std	0.9891 ± 0.0165
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

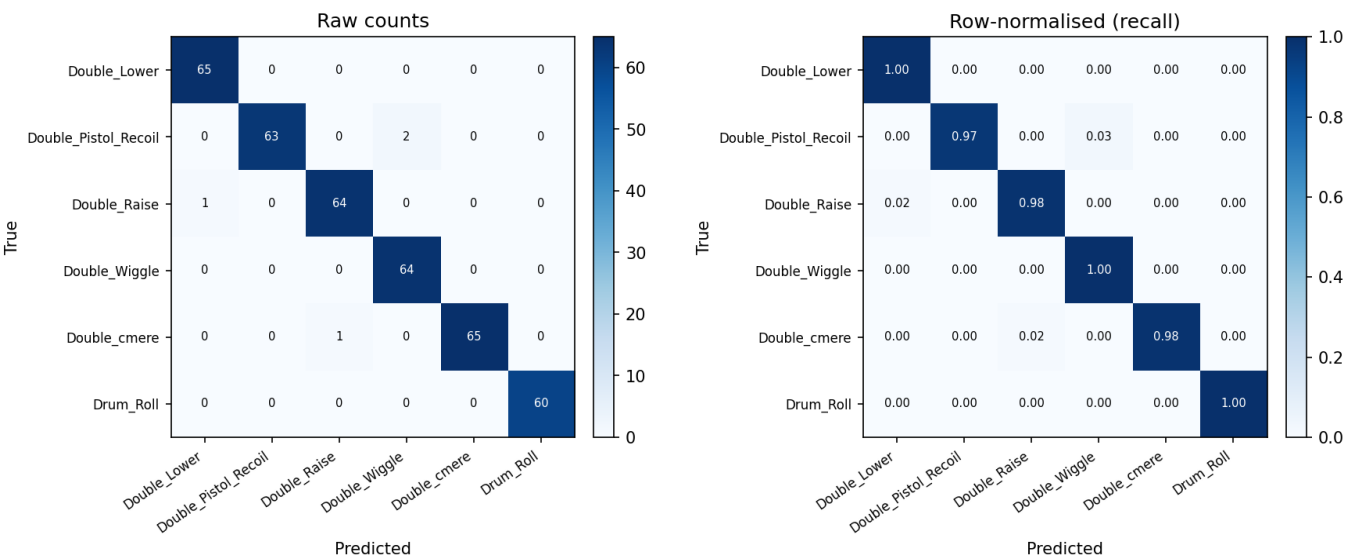
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
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2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	1.0000
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	0.9667
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	1.0000
13	8_Jestin	11_Mansh	331	24	0.9583

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	1.000	0.969	0.984	65
Double_Raise	0.985	0.985	0.985	65
Double_Wiggle	0.970	1.000	0.985	64
Double_cmere	1.000	0.985	0.992	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.990	0.990	0.990	385
weighted avg	0.990	0.990	0.990	385

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
LOUO mean ± std	0.9865 ± 0.0172
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

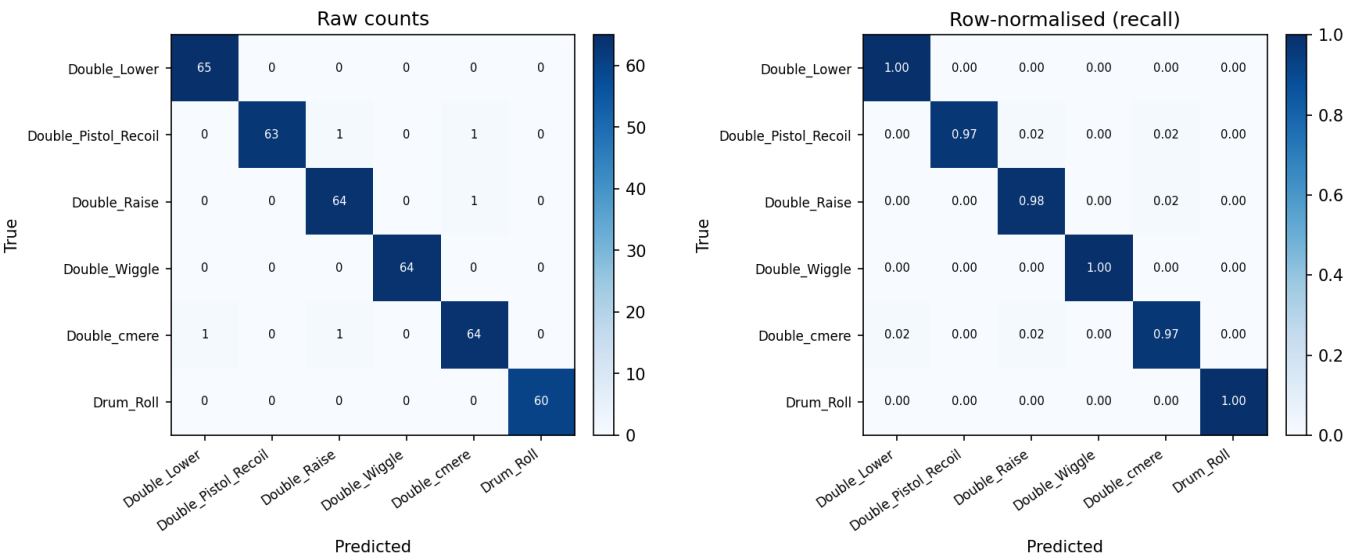
LOUO per-fold results

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9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.9583

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	1.000	0.969	0.984	65
Double_Raise	0.970	0.985	0.977	65
Double_Wiggle	1.000	1.000	1.000	64
Double_cmere	0.970	0.970	0.970	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.987	0.987	0.987	385
weighted avg	0.987	0.987	0.987	385

Confusion matrix (aggregated across folds)



Variant: BN_GAP_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

Trainable params	104,294
LOUO mean ± std	0.9788 ± 0.0553
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

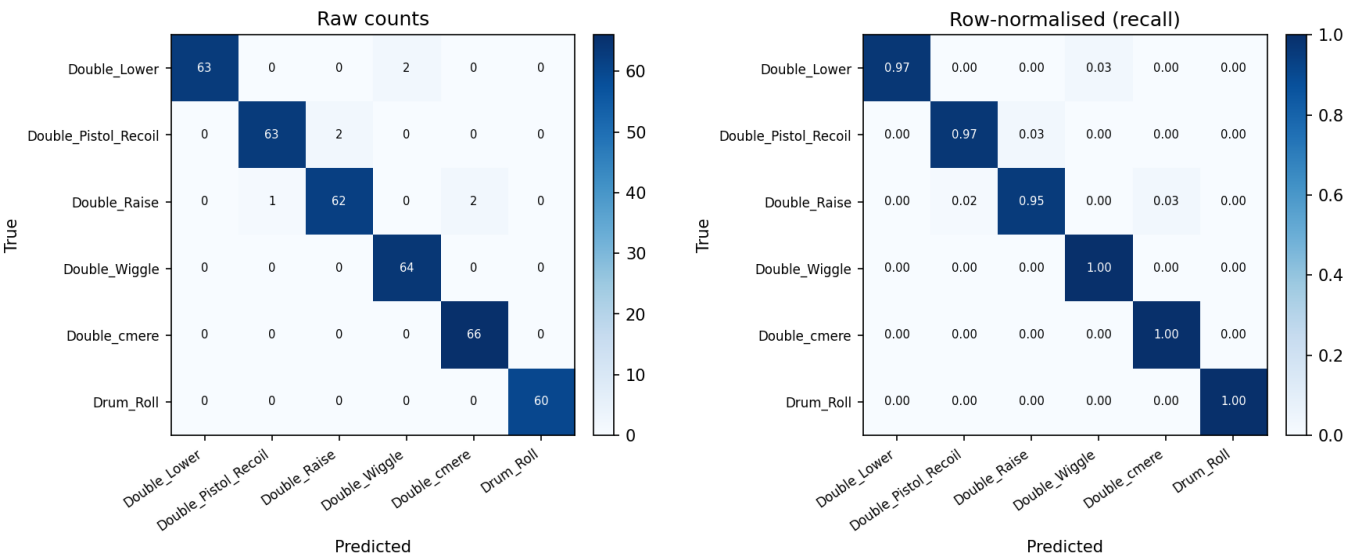
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	1.0000
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	1.0000
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	1.0000
13	8_Jestin	11_Mansh	331	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.969	0.984	65
Double_Pistol_Recoil	0.984	0.969	0.977	65
Double_Raise	0.969	0.954	0.961	65
Double_Wiggle	0.970	1.000	0.985	64
Double_cmere	0.971	1.000	0.985	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.982	0.982	0.982	385
weighted avg	0.982	0.982	0.982	385

Confusion matrix (aggregated across folds)



Final retrain (deployment model)

Variant	BN_GAP_Wide
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	385 samples (385 before augmentation)
Trainable params	104,294
Final train loss	0.0015
Final train accuracy	1.0000
Model path	— (not saved)
Labels path	— (not saved)
Scaler path	— (per-trial norm in use)